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THE BUSINESS VALUE AND INFLUENCE OF UX AND UI PRINCIPLES IN E-COMMERCE

BACHELOR THESIS

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Declaration

I hereby declare that I am the sole author of the thesis entitled “The Business Value and Influence of UX and UI principles in E-commerce”. I duly marked out all quotations. The used literature and sources are stated in the attached list of references.

In Prague on 11/05/2020

Signature

Valeriya Nekhymchuk

Acknowledgement

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Abstrakt

Tato bakalářská práce se zabývá významem principů uživatelské zkušenosti a uživatelského rozhraní pro webové stránky elektronického obchodu a hodnotou, kterou jejich aplikace přináší podniku.

V této práci byly definovány oblasti uživatelské zkušenosti a uživatelského rozhraní a rovněž byly představeny různé principy. Dále bylo popsáno téma elektronického obchodování se zvláštním zaměřením na to, jak se měří obchodní výkonnost webových stránek elektronického obchodu. Čtenář může očekávat, že bude seznámen s názvoslovím tématu.

Cílem bylo zhodnotit vliv principů uživatelské zkušenosti a uživatelského rozhraní na webové stránky elektronického obchodu. Cíl byl dosažen prostřednictvím průzkumů, které byly realizovány mezi lidmi se zkušenostmi s online nakupováním a mezi odborníky v oblasti elektronického obchodování. Další poznatky o tématu byly získány provedením osobních rozhovorů mezi odborníky, kteří se průzkumu zúčastnili.

Výsledky prokázaly, že odborníci, kteří navrhují a optimalizují webové stránky elektronického obchodu, a zákazníci, kteří je používají, považují zásady uživatelské zkušenosti a uživatelského rozhraní za důležité. Odborníci s tím souhlasili i při osobních rozhovorech a doporučili úzkou spolupráci mezi odborníky v oblasti designu, prodeje a analýzy dat.

Klíčová slova

User experience, UX, user interface, UI, e-commerce, ecom, business value, influence

JEL klasifikace

Abstract

This bachelor thesis explores the importance of User Experience and User interface principles to an E-commerce website and the value their application contributes to the business.

The fields of User Experience and User Interface design were defined and various principles presented. Furthermore, the topic of e-commerce was described with a particular focus on how an e-commerce website's business performance is measured. The reader can expect to be familiarized with the terminology of the topic.

The goal is to evaluate the impact of User Experience and User Interface principles on an e-commerce website. It was achieved by surveying people with experience in online shopping and experts in the field of e-commerce. Further insights on the topic were gathered by interviewing some of the expert participants.

The results demonstrate that User Experience and User Interface principles are highly valued both by experts who design and optimize e-commerce websites, and the shoppers who use them. The e-commerce professionals who were interviewed also agreed that User Experience and User Interface design bring value to an e-commerce business, and recommended close collaboration between experts in design, sales and data analysis.

Keywords

User experience, UX, user interface, UI, e-commerce, ecom, business value, influence

JEL Classification

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List of abbreviations

CTA	Call to Action
ISO	International Organization for Standardization
OECD	Organization for Economic Co-operation and Development
ROI	Return on Investment
RPM	Revenue per Mille
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience
WTO	World Trade Organization

Introduction

More and more retailers and brands shift to online, and some of them have never had an actual physical store in the first place. Understandably so, the time and cost of setting up an online business are significantly lower in comparison to the real-world alternative. As a result, today, countless e-shops sell everything from small goods like food and clothing to large goods like building materials and automobiles.

With so many things that come into play in online shopping, the look and the layout, existence, or non-existence of individual user interface elements may lead to users experiencing different emotions toward the website and further purchase or non-purchase. Therefore, it is crucial to take everything into account and understand how User Experience, along with User Interface design, can influence business retention indicators online.

Since User Experience and User Interface terms are getting more attention nowadays, the aim of this thesis is first to familiarize the reader with the given topic, to showcase the role of UX and UI when it comes to selling and buying online and most importantly, what are the effects that can be expected when paying more attention to website's user experience research and interface elements.

The goal of the theoretical part of the thesis is to provide a common understanding of e-commerce, what it is, how it works, and what surrounds it. Another goal is to provide in-depth information and thoroughly cover the terms of User Experience and User Interface, how they work together, and the difference between them. This part of the thesis also focuses on defining metrics that can often be used by online businesses to measure user experience and user interface performance.

The practical part of the thesis focuses on defining the role and business value of investing in UX and UI for an E-commerce website, which will be achieved by conducting a qualitative research in the form of a survey among online shoppers as well as e-commerce, UX, and UI experts about their observations on the current topic. In order to make sufficient conclusions the interviews with the e-commerce experts will be conducted to get to know their professional opinion and experience that involves User Experience and User Interface practices in their companies.

Defining the topic and motivation behind choosing

This bachelor thesis deals with the identification of business value and the role of UX/UI design in today's growing e-commerce industry.

The reason for choosing the topic is the author's great interest in User Experience, User Interface Design, and new technologies. Another reason is the author's belief that the Czech Republic has a significant e-commerce potential since it currently has the fastest-growing rate in Europe (Sedlák, 2020).

Thesis objectives

The primary and foremost objective of the thesis is to identify the importance of user experience and user interface principles in e-commerce.

That is done with the help of other secondary objectives of the thesis, which are:

1. To provide the theoretical background of E-commerce, User Experience, and User Interface areas.
2. To provide an overview of the E-commerce website and its performance measurement.
3. To research various metrics for measuring UX/UI performance.
4. To conduct and analyze surveys among the e-commerce consumers and experts, conduct interviews with the experts.

Expected results

The main contribution of the thesis is to introduce the reader to the concept of User Experience and User Interface design, their role in today's growing online world, shown on the example of e-commerce, the possibility to deepen the reader's knowledge of the topic and answer the research questions „What are the point that can be influenced by UX and UI principles in E-commerce?“ and the additional question “Is it worth for a business to invest in UX/UI design?”.

Thesis limitations

User Experience Design and User Interface Design, are relatively novel concepts when applied to e-commerce specifically. For this reason, no significant body of academic literature exists. The author has therefore relied primarily on articles from self-declared experts in the field of e-commerce, whether designers or business experts (e.g., Conversion Rate Optimisation specialists). In order to check the accuracy of these materials, the assertions contained within them were verified by those interviewees who took part in the survey.

Literature review

This subchapter is devoted to the review of university theses and resources that focus on User Experience, User Interface, and E-commerce. University theses and resources were found by using the keywords listed in the abstract. Both Czech and foreign literature in English will be analyzed in this part.

In order to elaborate on this thesis, a machine search of relevant sources was carried out with the help of a systematic search of keywords in the following professional sources:

- Final Theses Search in InSIS – the database for the submitted theses in the University of Economics. It allows searching theses by name, keywords, and author.
- Thesis.cz – gives access to numerous thesis across universities in the Czech Republic and Slovakia.
- Scholar.google.com – a powerful search engine powered by Google that allows us to look up various academic articles and research papers with the given keywords.

The following keywords have been selected for the final theses search:

- User experience
- User interface
- E-commerce
- Role
- Influence

Fröhlichová (2017), in her bachelor thesis *Redesign of user interface of a web application based on UX design principles* deals with the redesign of the web application and thoroughly describes UX, UI, and how they work together. The author mentioned various techniques that are used for the design stage of the website and covered web application testing principles. This paper covers a similar topic, which is helpful for the purpose of this thesis.

Filip Knopp (2015), in his diploma thesis *Increasing business performance of website* covers website business optimization with relation to user experience. It describes such metrics as Conversion Rate and Return On Investment, which will be highly contributing to this thesis.

Bachelor thesis *User Experience and use knowledge of cognitive science* by Blšáková (2014) very thoroughly describes cognitive psychology behind creating the websites, with the mentioning of concepts of User Experience and usability testing. In the chapter 'User Experience' and subchapter 'Business benefits,' she mentions how User Experience can contribute to a business's success, which will also be valuable for this thesis topic.

Another similar diploma thesis by Vaňková (2018) called *Cognitive aspects of "user experience" and web design* also covers psychological aspects of User Experience and web design. The practical part was performed with the help of a survey with questions in order to compare two websites' User Interface that included branding visibility, color schemes, readability, and overall usability as a whole.

Diploma thesis *A creation of a web design methodology with an emphasis on UX* by Hadamčík (2012) proposed his methodology with the different levels for creating web pages, that relies on User Experience. He also mentioned some aspects of the website User Interface that can have an impact on Conversion Rate.

For the purpose of finding the relevant books, the following keywords were used:

- Online business
- E-commerce

- User experience
- User interface
- Metrics

The book *Don't Make Me Think* written by Krug (2014) gives the reader an in-depth understanding of the basics of user-centric design. The book allows rethinking the already existing stereotypes about web design and helps to learn how to decide which problem to solve and how to correct it the right way. The essential elements of any web page are well described, why they should look like this, why they should be located here, and not elsewhere.

Web analytics demystified : a marketer's guide to understanding how your web site affects your business by Peterson (2004) thoroughly describes web analytics, understanding reports that need to be considered when running an online business. Since this bachelor thesis deals with the business value of User Experience and User Interface of e-commerce websites, this book is crucial for a deeper understanding of the topic of website performance measurement.

Cost-justifying usability: an update for an Internet age book by Bias and Mayhew (2005) covers the topic of everything related to cost-justifying usability and User Interface Design's Return of Investment. It describes different aspects of usability, including the website's, which is relevant for the purpose of this thesis. Most importantly, it emphasizes how closely the User Experience and User Interface work together (Bias, Mayhew 2005).

Designing the Obvious: A Common Sense Approach to Web Application Design by Hoekman (2011) comprehensively provides an excellent framework for approaching User Interface design. The dedication at the beginning of the book says: „This book is dedicated to anyone who has ever used a Web application and resented the experience.“ This gives the idea of what this book is focused on: User Experience. Each chapter in the book can be seen as a separate design principle and provides numerous practical tips. Does not contribute much to the topic of the thesis directly but gives a better understanding of what works in web application and what does not.

A significant amount of information about the field of E-commerce was provided by Suchánek (2012) in the book *E-commerce: e-business and e-commerce concept*. It gave an in-depth understanding about multimedia, e-commerce, e-business, the detailed description of e-commerce website and its elements, which is crucial for the purpose of this thesis.

1 Definition of the primary terms

1.1 The definition of User Experience (UX)

“You cannot understand good design if you don’t understand people.”

– Dieter Rams, Product Designer of Braun

According to International Organization for Standardization 9241-210:2010 (2010) standard, UX can be defined as a *“person’s perceptions and responses resulting from the use and/or anticipated use of a product, system, or service.”* It also includes all kinds of users’ emotions, preferences, and feelings that form before, during, and after use.

Donald Norman, one of the main experts in the industry, (2013) explained UX in comparison to opening a door, where a user instantly knows whether the door needs to be pushed or pulled.

User Experience represents the consistency of aspects such as marketing, engineering, and design, with which the user interface blends. Many definitions of UX state that between the main conditions of excellent user experience are mainly comfort, which should meet the needs of the target customer without problems or difficulties. This is linked to other conditions, which are the simplicity and elegance with which users enjoy the product or service (Norman and Nielsen, 2016).

Jesse James Garrett in his book called *The elements of user experience* (2011) describes UX as *“the experience the product creates for the people who use it in the real world”* and further points out that it is a crucial part of all kinds of products and services, in-real-life and online.

In other words, we can define UX as emotions and feelings a person has in the process or after interacting with a specific object (product).

1.2 The definition of User Interface (UI)

The ISO 9241-110 defines UI as “all components of an interactive system (software or hardware) that provide information and controls for the user to accomplish specific tasks with the interactive system” (ISO 9241-110, 2006).

In layman’s terms, the user interface is everything that is designed into an information device with which a person can interact. The UI includes not only the way users interact with websites and applications, but also the screen, keyboard, mouse, or desktop appearance (Rouse, 2019).

What users see in the process of browsing the webpages is often called Graphical User Interface, which came into existence after the first type of UI - Command-line interface, but along with the graphical elements. (Rouse, 2006).

To summarise, UI is part of a computer system that serves to communicate a user with an application or technology device and, in many cases, also provides information transfer between multiple users and a device. When it comes to human-computer interaction, UI serves as a mediator in this process.

1.3 Other definitions used in UX/UI field

Interaction Design (IxD)

Interaction Design (IxD) focuses on creating products that enable the user to achieve their goals in the best possible way. The user experience is formed through interaction with the interactive product. This interaction is essentially a dialogue where the user's system of action responds to the product's action. This is the feedback that the user performs the following action. At the end of the product, user action and feedback chain should be the desired user target. Interaction design deals with the design of these interactions. It solves user needs and requirements (based on them defines the functionality of the product) and looks for the optimal way to satisfy them (Siang, 2018).

There were defined five dimensions of IxD (Silver, 2007):

- **Words** are the dimension of the appropriateness of using specific terms for the desired purpose and overall consistency. The overall impression of "Tone of voice" should match the character of the product.
- **Visual representation** represents all visual manifestation. It forms an overall structure that is achieved by the appropriate interplay and includes, for example, pictures, typography, diagrams and icons, and forms an overall structure that is achieved by appropriate interplay.
- **The physical form** defines spatial characteristics. Image and proportions can be a natural bearer of information on the purpose and use.
- **Time** describes that the product system changes overtime during the interaction. The time dimension, for example, brings sound or video effects and animations.
- **Behavior** is a dimension of actions and reactions. The first three dimensions allow interaction, and the fifth dimension of behavior in the fourth dimension of time defines its form.

Visual design

According to Alice Twemlow, the author of the book called *What graphic design is for?* described Visual design as an interdisciplinary field that uses expertise from many areas such as typography, animation, writing, photography, marketing, or psychology. Over time, niche fields such as broadcast design, sound design, game design, editorial and book design, corporate identity design, and interactive design have been identified. Graphic design is still

evolving. Economic and cultural-social requirements and processes influence it. (Twemlow, 2008)

In other words, visual design has a significant impact on the user experience, which is further enhanced by the initial contact with the product during the formation of the first impression. Eye-pleasing aesthetics can attract and retain attention, while an unpleasant, disharmonic appearance can discourage the user before any further interaction occurs.

Customer Experience (CX)

The relevant definition of CX is provided by Gartner, one of the leading research and advisory companies, states that it is "the customer's perceptions and related feelings caused by the one-off and cumulative effect of interactions with a supplier's employees, systems, channels or products." (Gartner, n.d., a)

Customer experience management is the design and implementation of company interactions with customers in such a way as to meet or exceed customer expectations and thereby achieve greater satisfaction, loyalty and degree of identification with the company (Gartner, n.d., b).

In other words, from these any many other definitions it is defined as emphasis on the client's overall experience and focus on the whole life cycle of the relationship between the company and the customer, not on separate partial interactions and transactions. From this and the following scheme we can conclude that UX is one of the part that create CX.



Figure 1: CX including UX (Source: Coding Infinite, 2019)

1.4 The definition of E-commerce

E-commerce

Cambridge Dictionary defines the term as “the buying and selling of goods and services over the internet” (Cambridge Dictionary, n.d.).

This definition helps us understand that it is considered to be a process. The following definition also confirmed that. “E-commerce is understood to be a process of buying, selling, distributing and exchange of products, services, and information using computer networks, including the Internet” (Gála, 2009).

World Trade Organisation (n.d.) definition of the term is understood to mean “the production, distribution, marketing, sale or delivery of goods and services by electronic means.”

From the Czech academic literature, it is worth mentioning Suchánek’s (2012, p.11) definition of e-commerce, which states that “E-commerce can include all business activities and services during which data transfer is directly carried out using electronic tools directly related to the realization of business cases, and in the case of purchasing software or other information, these can also be transferred electronically to the consumer.”.

Many times, e-commerce gets oversimplified and understood as buying and selling over the Internet and platforms such as Amazon. However, all electronically-mediated transactions between the organization and any third party should be included when defining e-commerce. Non-financial transactions, such as customer support and requests for additional information, are also considered as part of e-commerce (Chaffey, 2019).

The most significant part of e-commerce is mainly e-shops. They may also include any website on which specific products or services are offered and also allow their order, for example, by email. (Adaptic, n.d.).

E-business represents the highest level for business activities that are carried out using information and communication technologies. If we focus on purely commercial activities, we can talk about e-commerce. In the field of e-commerce, one of the vital communication interfaces is Internet shops (e-shops). (Suchánek, 2012)

For the purposes of this paper, unless explicitly stated otherwise, e-commerce (EC) will be delimited by a narrower definition by Gála. The terms e-commerce in the broader sense and e-business will be considered synonymous.

E-shop

E-shop, electronic shop, or sometimes also an online shop, is a particular web application (web portal) serving as a communication interface of e-commerce. It enables the presentation of goods and order realization in a virtual environment. E-shops are understood as a subset of e-commerce. They are similar to brick-and-mortar shops and have

their URL2 address, which indicates the place in the network where sellers and shoppers can conduct business activities (Suchánek, 2012).

It is the most common model for e-commerce. As an example, we can imagine an Amazon website worldwide and Alza on the Czech market. These are companies that do not produce goods but only sell them. (Machkova, 2014).

2 User Experience and User Interface

For better understanding of the purpose of this thesis, it is crucial to familiarize the reader with the fundamentals of User Experience and User Interface processes. The following chapter will provide an in-depth overview of UX and UI principles.

2.1 The relevance of User Experience

User Experience has been a topic of discussion over the last year and over the past fifteen years the interest in the field of User Experience has grown significantly.



Figure 2: UX interest over time (Source: Google Trends, 2020)

As for the future, the growth is going to get even more rapid, as it was predicted by Nielsen (2017) that by the year of 2050 the number of UX professional around the globe will reach a hundred million, which is 1% of the world population.

UX is still in its infancy stage, yet we can see in the graph above how fast it grew. It is true that a certain proportion of people who put themselves before the name of their UX position actually do not work with the user experience, or even little. These are graphic artists who paint UX in their heads but do not really do any analysis or verification of hypotheses, which does not correspond to the description of their work in the UX field.

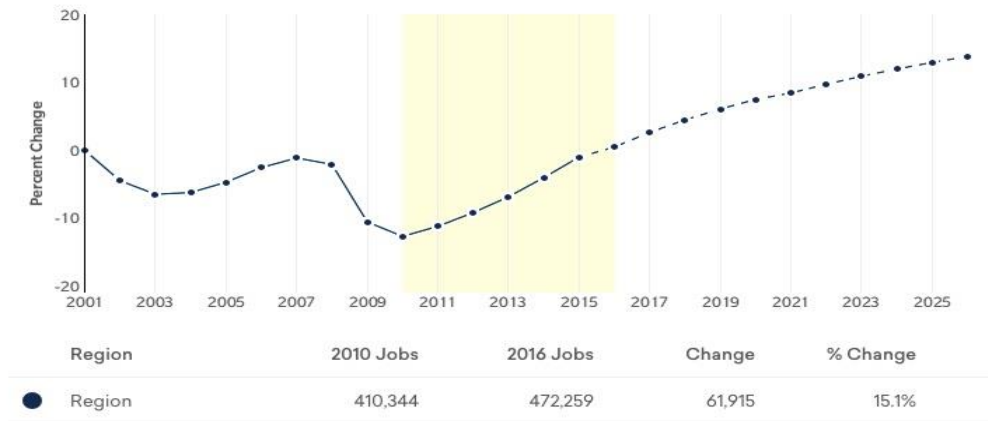


Figure 3: UX jobs over time (Source: Emsi, 2007)

Emsi has also confirmed the fact of significant growth of demand for UX/UI designers in the Information Technology industry and agree that for the foreseeable future the growth is going to continue (Emsi, 2017).

2.2 UI as a part of UX

UI and UX are highly-discussed terms nowadays and most of the time are mentioned in relation to each other. In previous subchapters they were defined separately, but it is vital to point out how the two disciplines are commonly seen in the design world.

As stated by Kirmaier, UX consists of various parts. From the following scheme, we can see that UI is a subset of UX, however, however it does not make it any less important (Kirmaier, 2019).

Torre in the article for UX Collective stated that *“The user experience is the result of their interaction with a product or service, which means that user experience includes, but is not limited to, a user interface. The UI is one of the contributing factors for the end user experience”* (Torre, 2019).

In other words, from the e-commerce point of view, the user first interacts with the User Interface elements on the website and User Experience is an emotional effect during and after this interaction, so it is important not to separate the terms and understand why they are mentioned together in this paper.

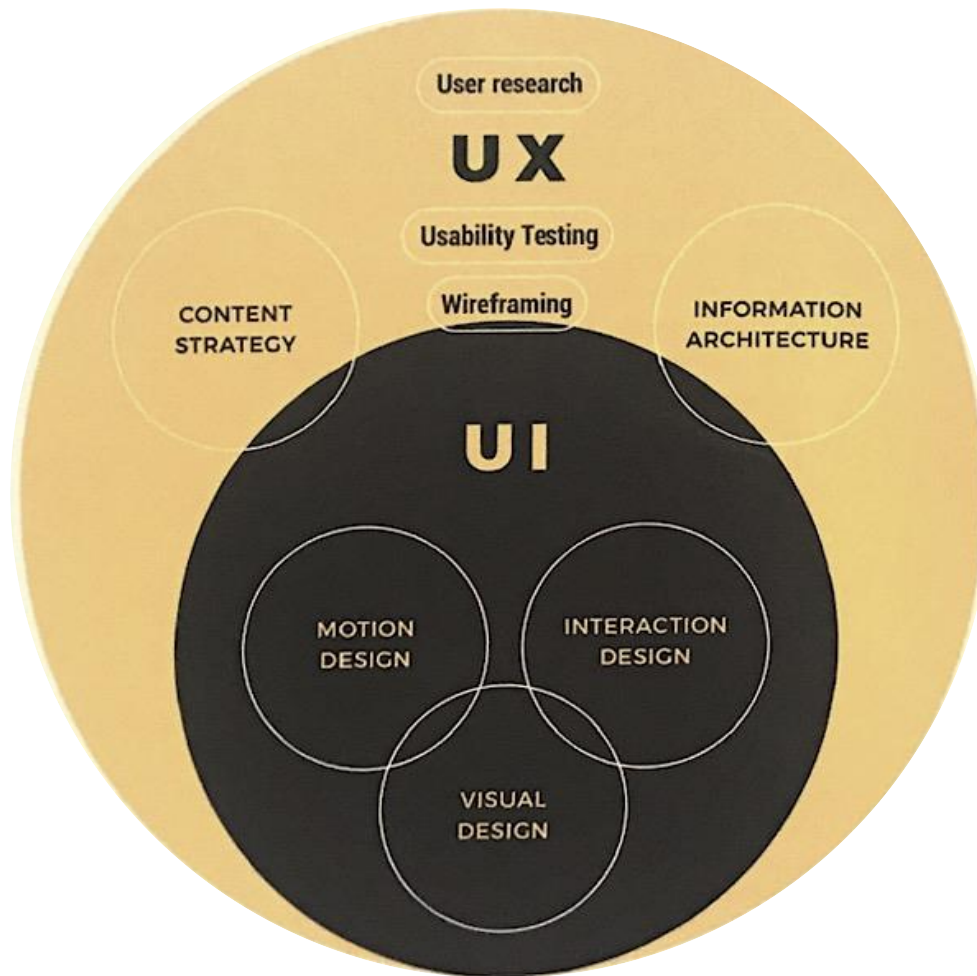


Figure 4: UX design umbrella (Source: Kirmaier, 2019)

2.3 The principles of User Experience

This subchapter is dedicated to defining UX elements or fundamental parts. Since there is no single distinct group of principles, the author decided to introduce two groups of those that would give the reader a better idea about the UX and how it is about in practice.

User Experience Design is a relatively new field, so there is no single commonly-agreed upon a set of principles that defines it. Businesses might have their internal methods and practices when it comes to UX. Therefore, the author has chosen two sets of principles that are useful methods for breaking down the design of user experience, in particular to a digital shopping experience. When discussing the application of UX principles, it is therefore assumed that the current or intended customers are already fully understood in terms of their demographics and their needs.

UX honeycomb

Peter Morville, an expert in information architecture and UX, lists a number of aspects that shape the resulting user experience. For a product or website to provide positive user experience, it must meet the following 7 aspects (Morville, 2004):

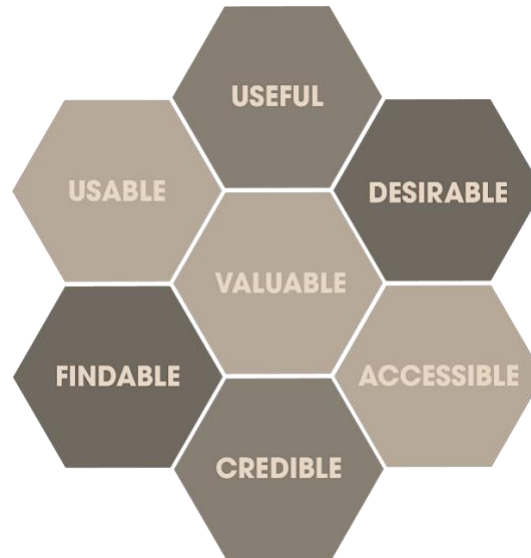


Figure 5: UX honeycomb (Source: Morville, 2004)

- **Valuable** (Is it valuable?) – Morville points out that the products are considered to be valuable if its features are designed in a way that supports user's wants and needs.
- **Usable** (Is it easy to use?) – Usability is a vital part of product design. Things such as findability, readability, learnability, and overall consistency are considered to be a part of usability.
- **Findable** (Is it easy to find?) – The website navigation should be easy, and users should be able to find what they need.
- **Desirable** (Is it fun and engaging?) – This element describes the emotional aspects such as the visual and functional appeal of the website.
- **Accessible** (Is it accessible to all people?) – The website should be accessible to all people that are using it, including people with disabilities.
- **Credible** (Is it trustworthy?) – The website should contain the elements that indicate that a user can trust what is shown on the website and make sure his data is in good hands.
- **Useful** (Is it useful in any way?) – The more useful the website it, the better the experience.

At the center of this model is the main development criterion - the value of the product. As a rule, the business owner is interested in ensuring that his product is valuable to the intended users, and also has sufficient competitive ability among a number of similar applications.

In addition to the ratio of all criteria to the central one, it is important not to lose sight of how the indicators of the outer circle border each other. They are arranged in such a way

that adjacent criteria influence each product definition and each of them is necessary, but not sufficient on its own. Therefore, this model should be used only in its entirety.

The advantage of this model can be considered that it fulfills several essential goals at the same time. Firstly, it is very convenient to identify priorities by selecting them from the proposed UX quality cells and not forgetting to use the others that affect the selected ones. Secondly, this model supports a modular approach to the development of web design.

Thirdly, if we consider the developed experience of interaction through each cell, it becomes easy to analyze whether everything has been done so that this quality takes the maximum value required by the customer.

Five UX planes

Garret (2011) broke UX down into Five UX element planes.

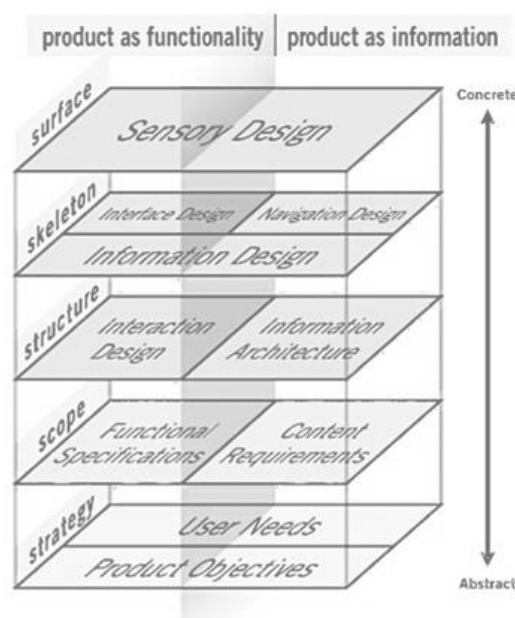


Figure 6: Five UX element planes (Source: Garret, 2011, p.22)

Interaction experience is the result of many large and small design decisions on how the product looks, how it behaves, and what actions it allows to perform. These decisions, based on each other, form the experience of interaction and affect all its aspects. (Garret, 2011)

According to Garret (2011), if the following five user experience elements are executed continuously, they can bring positive effects to the developed products by meeting business goals and satisfying the customers and develop their loyalty (Garrett, 2011):

- **Strategy level** – Describes the necessity to understand what the audience wants and how these wishes are consistent with its other needs. This level also includes the business goals of the company.
- **Scope level** – When it comes to the software part, the strategy is transformed into a set of capabilities by creating a functional specification - a detailed description of the process's functionality. In the information part, capabilities are equivalent to

content requirements - this is a description of the various content elements that have to be created.

- **Structure level** – According to Garret, is where user experience is formed. At the software half of this level, capabilities acquire a structure through interaction design, during which we determine how the system will behave in response to user actions. In the information space, the structure is determined by information architecture.
- **Skeleton level** - On both halves of the layout level, information design is used - presenting information in a form that facilitates its perception. For software products, the layout also includes an interface. The decisions made during this stage can lead to good or bad UX.
- **Surface level** - The visual design is responsible for the appearance of the final product. UX design decisions made at this stage must strengthen and support all decisions regarding other elements of user experience.

Everything in the field of User Experience relates to designing for particular sets or types of users. These groups of users, often referred to as user personas, vary by their personal characteristics (e.g., demographics) of their needs (e.g., what might drive their interest in a particular product) (Kirmaier, 2019). Therefore understanding the types of user personas for whom the experience as intended is paramount.

While user research is a fundamental part of UX design, to allow us to focus on the impact of User Experience principles on e-commerce, in this thesis, the author has chosen not to detail the user research process.

2.4 The principles of User Interface

10 Usability Heuristics for UI design

In 1994, Jakob Nielsen, an expert in human-computer interaction, described ten heuristic user interface principles. It is a set of recommendations that make it easier to use computer applications (Nielsen, 1994):

1. **Visibility** - The system should provide feedback to the user on where he is and what is happening.
2. **Language** - The system should communicate with the user in natural language.
3. **Control and freedom** - The system should offer the user a choice and return.
4. **Consistency** - the system should contain elements that are familiar and readable to the user while maintaining consistency throughout the application.
5. **Error prevention** - the system should undergo extensive testing to eliminate errors in application functionality.
6. **Support recognition** - the system should contain elements to indicate what it has done or where it is located directly on the current page.
7. **Flexibility and efficiency** - the system should offer the possibility to speed up some frequent actions for experienced users.
8. **Aesthetics and minimalism** - the system should contain only the necessary amount of information and avoid excessive text.

9. **Clear error messages** - the system should inform the user of any bugs or errors and offer the option of correction or return.
10. **Help and documentation** - the system should contain concise and understandable answers, such as the FAQ (Frequently asked questions) section.

Gestalt principles

Gestalt principles, or gestalt laws, founded by German psychologists Max Wertheimer, Wolfgang Kohler, and Kurt Koffka and is focused on how people interpret the world. The whole idea of these principles is that the “whole of something” is more noticeable to us than the details (Todorovic, 2008).

The following principles can also be applied to the area of UI and are commonly used in web design practices (Babich, 2019):

1. **Similarity** – this principle says that objects with similar characteristics, similar color or shape, or similar sizes - are determined by the human brain as a group of elements. When it comes to UI design, it can be used to create similarities between sizes, fonts, or shapes.

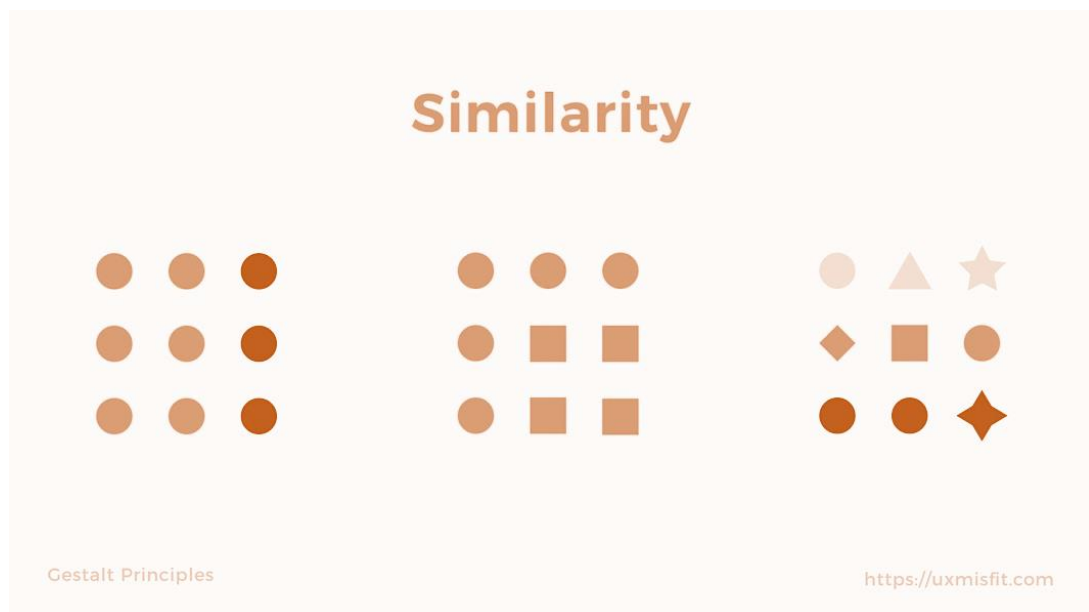


Figure 7: Similarity principle (Source: Uxmifit.com, 2019)

2. **Proximity** – this principle claims that objects that are located next to each other are perceived by the brain as groups. From the UI perspective, we can see that circles on the left side form a separate group to the group of circles on the right side.

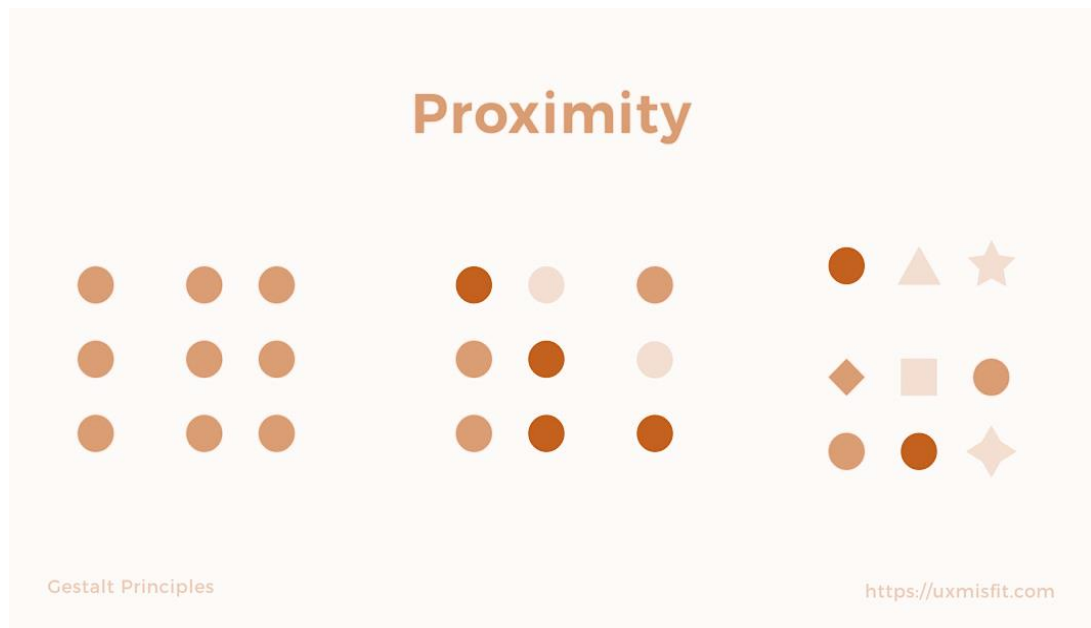


Figure 8: Proximity principle (Source: Uxmifit.com, 2019)

3. Closure - this law states that our eyes, mainly visual perception, group things that can be combined with each other. In the following image, as an example, it reflects the tendency to complete the figure so that it takes on a full form.

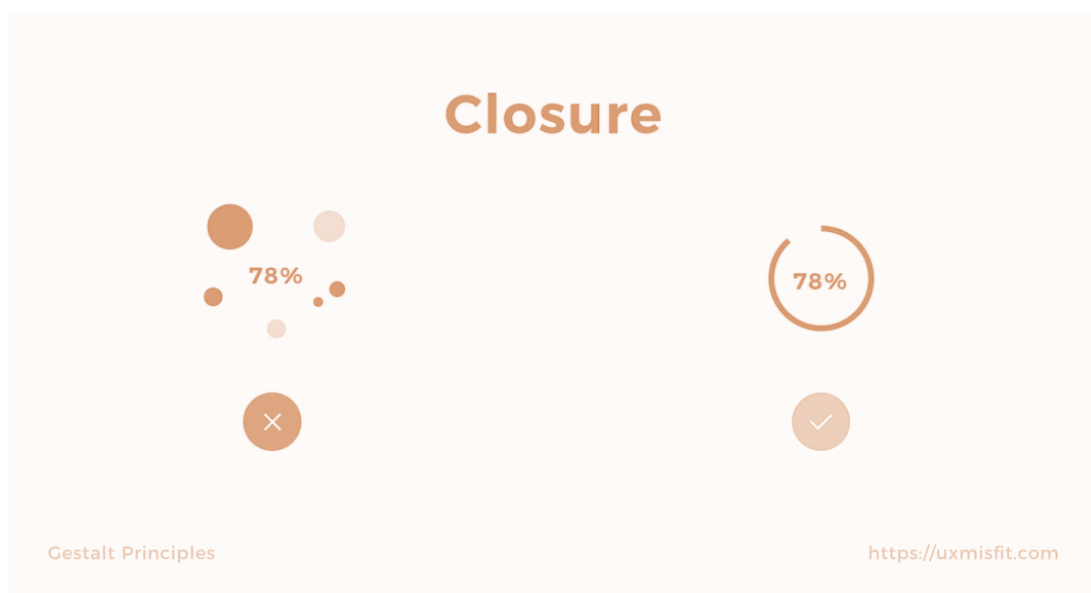


Figure 9: Closure principle (Source: Uxmifit.com, 2019)

4. Continuity – claims that objects organized in lines or curves are also perceived to be grouped.

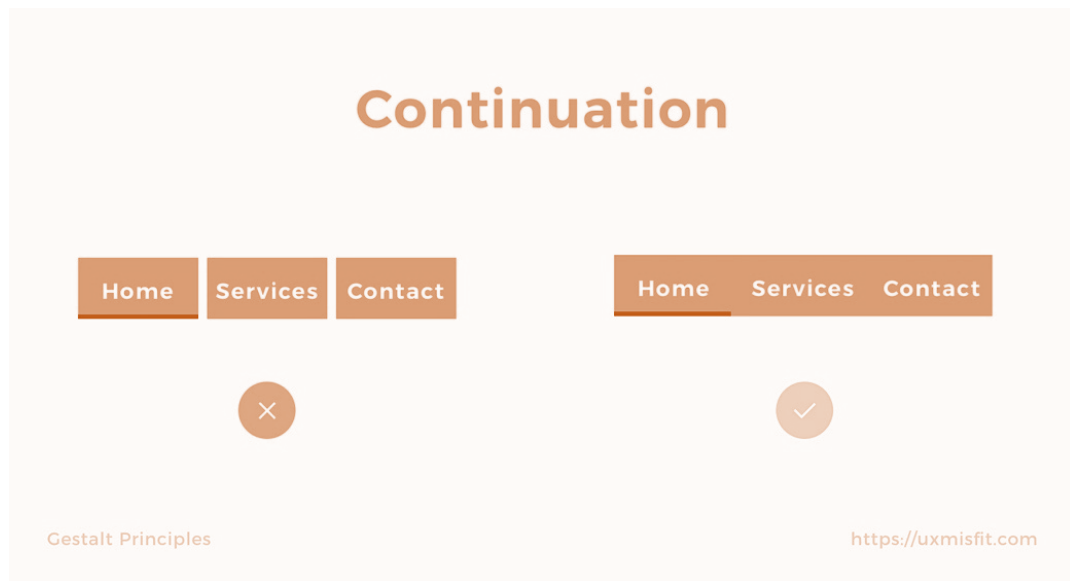


Figure 10: Continuation principle (Source: Uxmifit.com, 2019)

5. Figure & Ground – describes how we perceive and separate visuals into levels or layers.

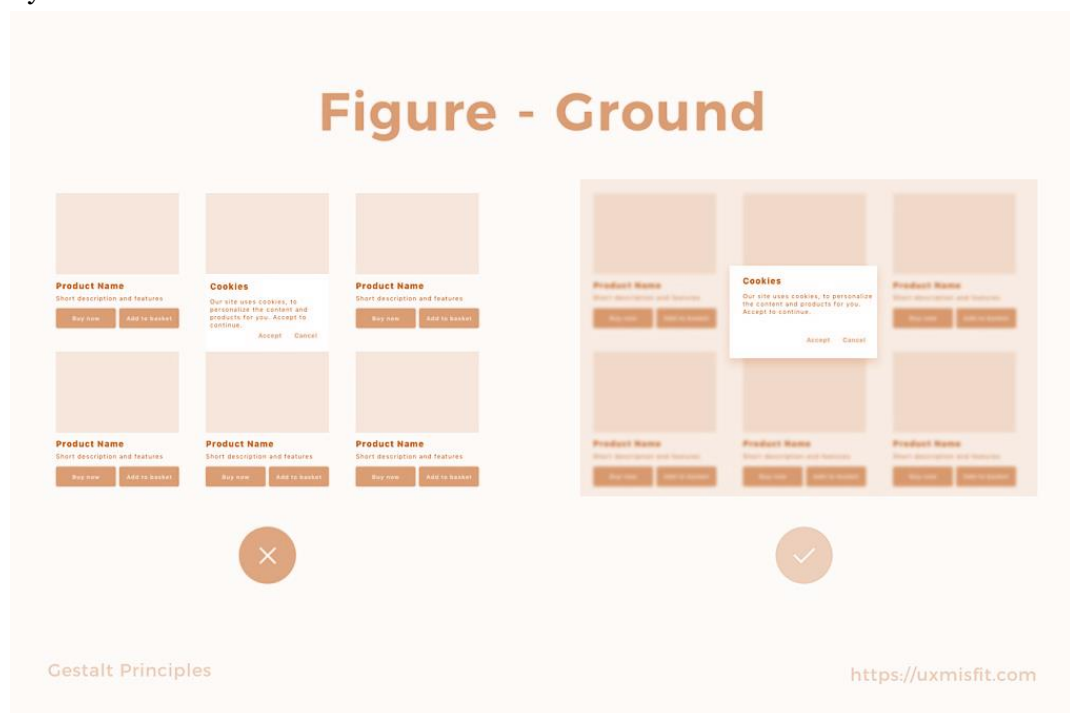


Figure 11: Figure-Ground principle (Source: Uxmifit.com, 2019)

6. Symmetry – states that our mind gravitates towards symmetrical objects, because symmetry gives the impression of stability and order.

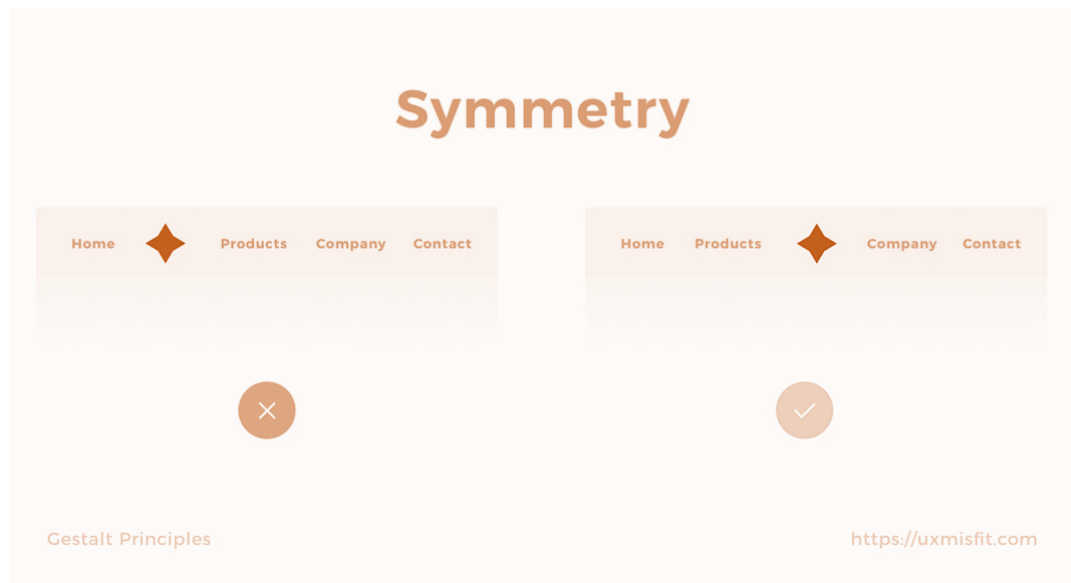


Figure 12: Symmetry principle (Source: Uxmifit.com, 2019)

3 E-commerce website

3.1 The elements of e-commerce website

This subchapter is focused on identifying all the essential parts of a regular online e-commerce website from the customer point of view. It describes the elements of an online shop from the customer's (user's) perspective.

Věra Mulačová, in the book *Business in the 21st century* (2013), described the following elements as one of the most common and must-have on e-commerce websites. Van Duyne, Landay and Hong (2005) add that one of the most frequently used and important elements of any website, including e-shops, is a Search bar. Therefore, it will also be included.

Table 1 E-commerce shop fundamentals (Source: Mulačová, 2013, p. 114, Douglas, 2005, p.530 and Sedlacek, 2006))

Element	Characteristics
Home page	The home page contains the primary business and contact information. This is the first page a customer sees when he enters the website.
Catalog of products and services	Catalog of products and services offers the possibility of a closer display of the product and description of detailed information.
Search bar	A graphical element that represents an input field that enables the user to look up a desired item or information using key-words.
Shopping cart	A shopping cart represents a virtual place for purchased products or services.
Delivery method	This page helps the customer to choose the preferred delivery and shipping method.
Checkout page	The checkout page offers the customer different options on how they want to pay. Usually, they choose between bank transfer, credit card, or cash on delivery.

Home page

A homepage is the first page of an online store where a user should know what kind of online store it is and what goods or services the store offers. This page represents the whole e-shop. Its role is to arouse interest in a potential customer and evoke a desire to obtain further information about an e-shop product or service. It is primarily accessed by users searching for a particular store under its name or the product category it offers. The fundamental elements that appear on this page are logo, magnifier - text search option in the store menu, shopping cart icon, and icon for user registration/login. (Sedlacek 2006)

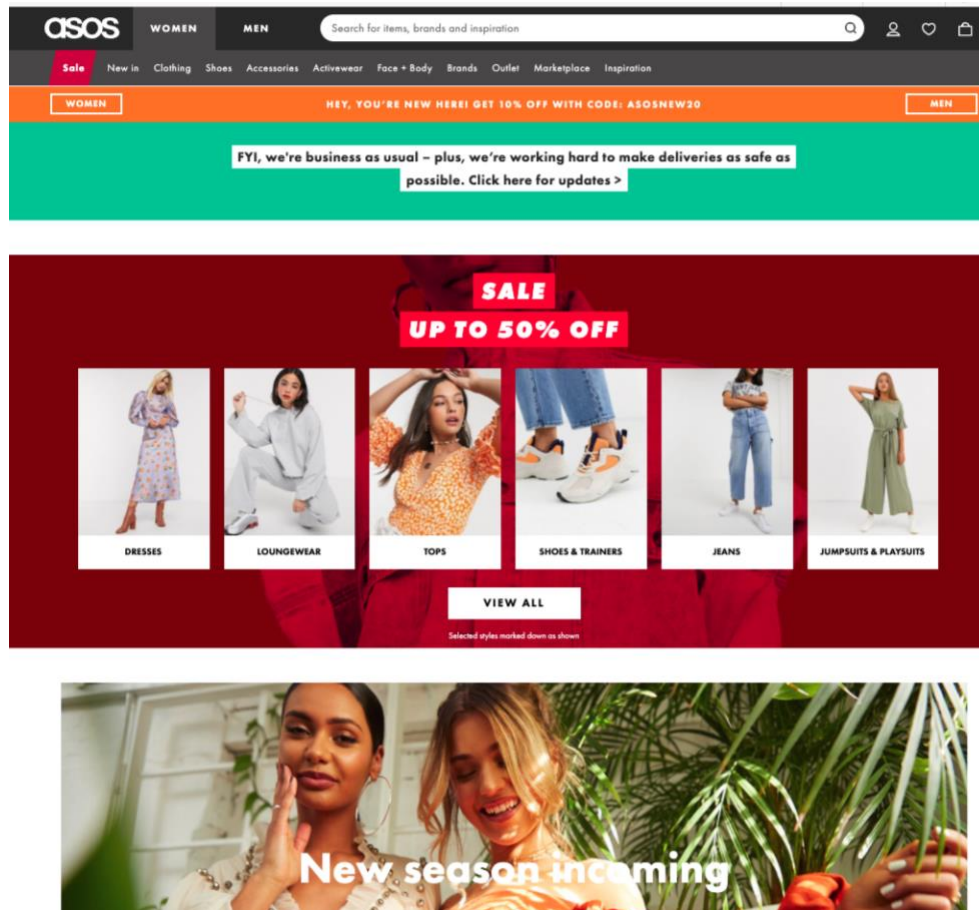


Figure 13: Asos homepage (Source: Asos.com, 2020)

Catalog of products and services

The catalog of products or services represents the company's offer. On its pages, the user will find more detailed information such as price, description of the parameters, and a photo of a particular product. The user should easily, quickly, and transparently provide the information they are looking for (Sedlacek 2006).

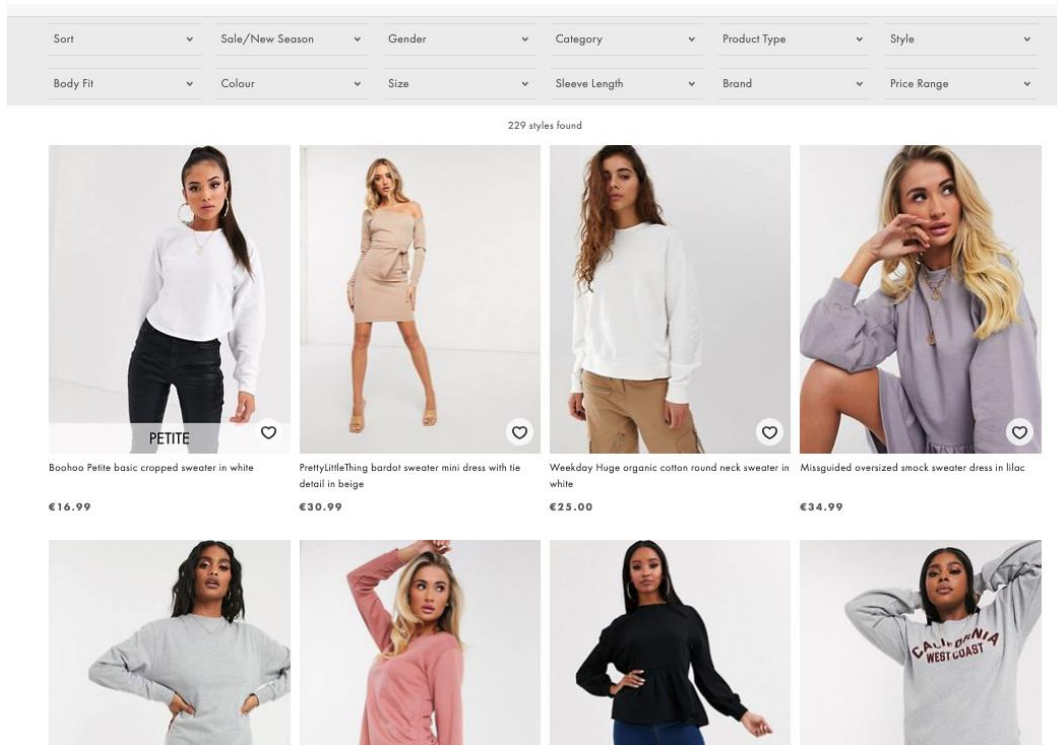


Figure 14: Asos product catalog (Source: Asos.com, 2020)

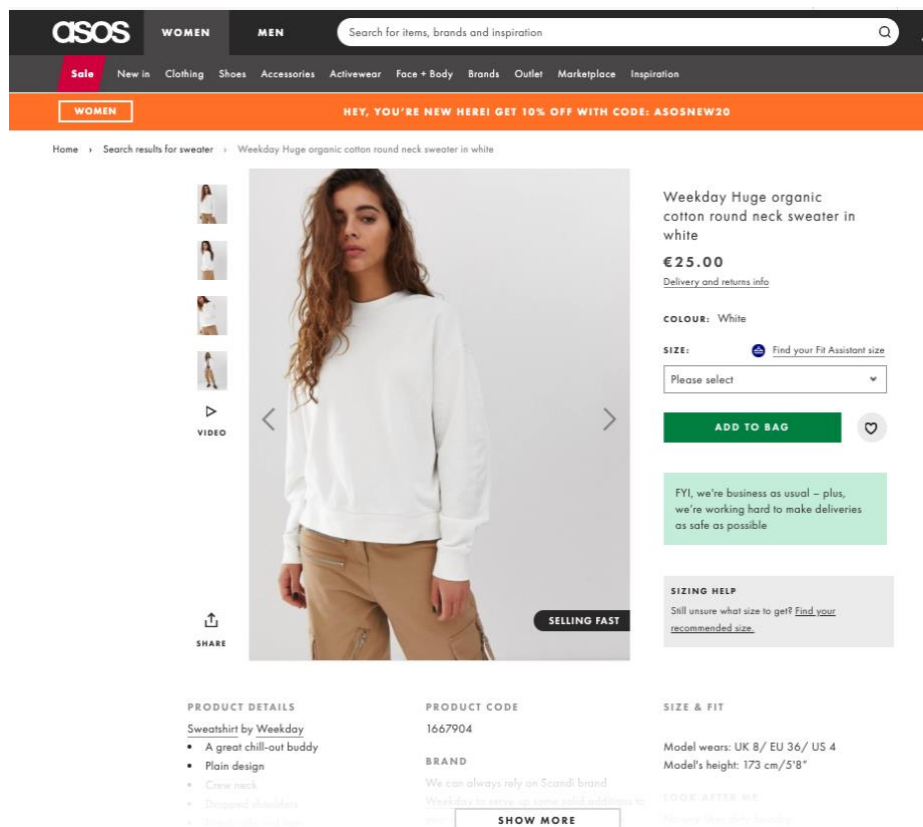


Figure 15: Asos single product page (Source: Asos.com, 2020)

Shopping cart

Just as in the physical form of the shopping cart, as well as in the virtual one, the customer collects the products he/she has chosen in the e-shop and wants to buy them. An unwritten rule of user-friendliness is placing the cart in the site header, usually to the right. The basket should always be in the same place, be visible, and accompany the customer throughout the purchase. The shopping cart feature should also include interaction when a user places a product in it and gives them an overview of the quantity and price of the products they embed (Sedlacek, 2006).

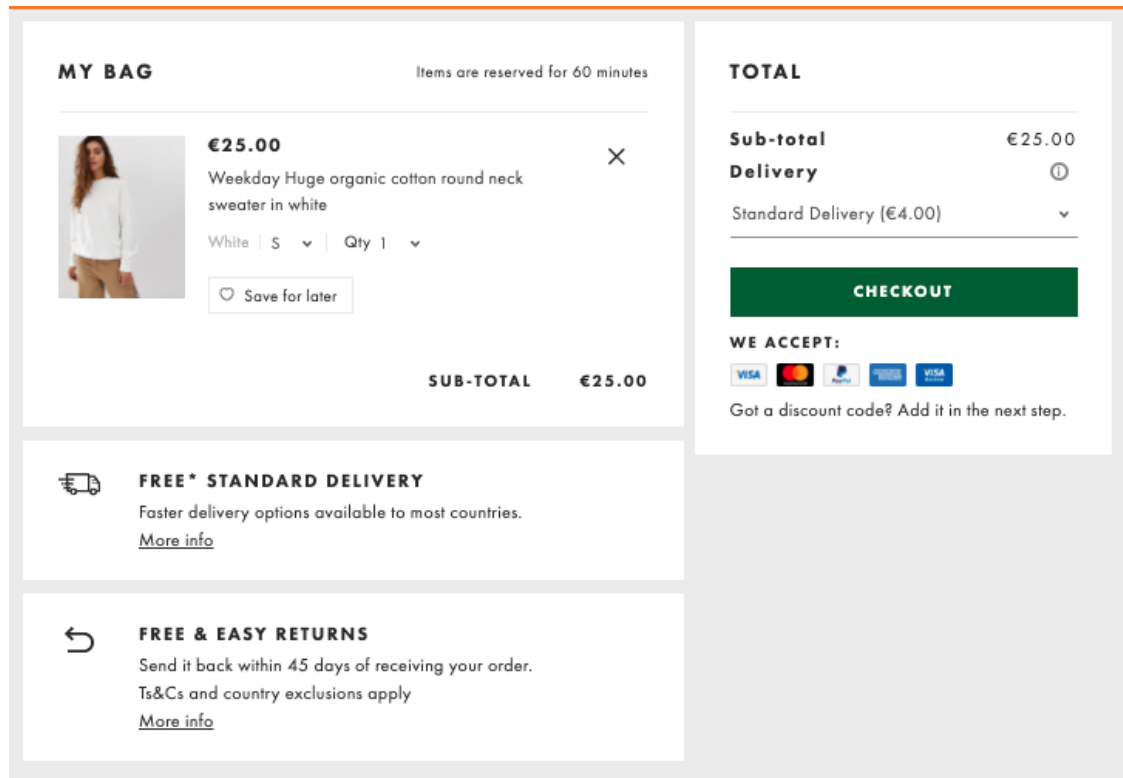


Figure 16: Asos shopping cart page (Source: Asos.com, 2020)

Checkout page and delivery method

The shopping process usually ends after checkout. The customer finally confirms the selected goods at the checkout and chooses the method of payment for the purchased goods. Each e-commerce website mediates different payment options, which may also be related to the way the items are delivered (Suchanek, 2012).

Figure 17: Asos checkout page (Source: Asos.com, 2020)

From the few examples of the e-commerce website, we can conclude that the overall design, flow, and various elements, as well as information, is made according to most of the UX and UI principles mentioned in the previous chapter.

From the UI perspective, there can be observed all of the Gestalt principles that are similarity, proximity, continuation, and symmetry.

From the UX point of view, we get all the necessary information from the system, which makes it useful, and the website is easy to use. In the checkout page, we can see appropriate labels that suggest that the website can be trusted when it comes to the private information like the address and credit card number, which implies that the credibility aspect is in place.

3.2 Customer journey

Customer journey is a step-by-step channel a customer should ideally go through in the purchase process (Binder, 2014).

The following scheme describes the ideal process of the customer journey on an e-commerce website. The scheme is further divided into three-part, which are Awareness, Consideration, and Acquisition.

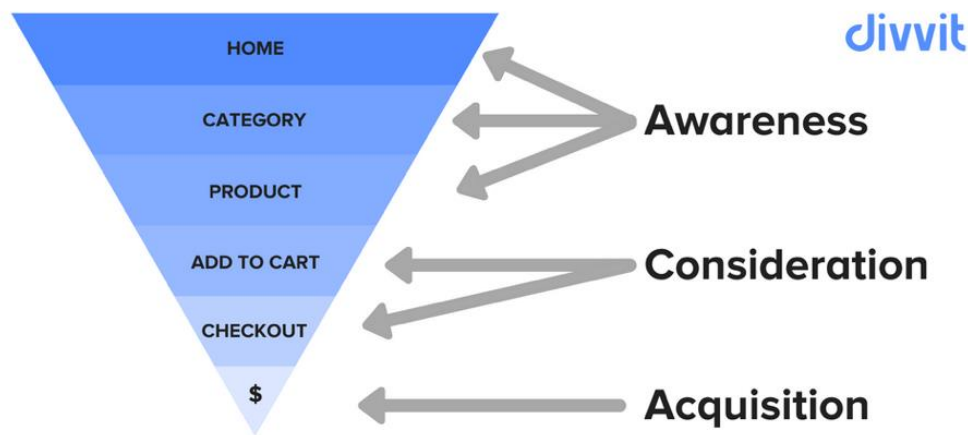


Figure 18: E-commerce customer journey (Source: Blankenship, 2018)

Awareness part includes the Home page, Categorisation on the website, and Product pages. At this stage, the customer is introduced to the website, and these elements help him understand what the brand is and what it can offer. Consideration part consists of the “Add to cart” action and Checkout process. At this stage, the customer goes through the longer evaluation process on whether or not he is going to purchase your products or services, and the customer engagement plays a crucial part in this decision-making process. Lastly, the Acquisition part is where the customer decided to purchase on the given website (Blankenship, 2018).

There are, of course, many different aspects that come into play when a user is visiting the website and many reasons why he would not make a purchase. These exceptions will be further discussed in the next chapters.

3.3 E-commerce website performance measurement

Measurement and research are of an e-commerce website that is crucial to evaluating how well the business is doing.

There are a number of metrics that are used for measuring e-commerce performance and can also be related to how well designed the website is. In an e-commerce world, web analytics is one of the tools to access the numbers.

The definition of web analytics from the Web Analytics Association (WAA) states: "Web analytics is measuring, collecting, analyzing, and reporting Internet data to understand and optimize web usage" (WAA, n.d).

According to Stebbins, web analytics provides insight into the functioning and productivity of the company's presence on the Internet. That is, a look at the web page itself, advertising and referring web pages. (Sebbins, 2007).

Conversion rate

Conversion rate is a ratio of those who complete the desired action (e.g., order goods) to all visitors and is given as a percentage. (Adaptic, n.d)

$$\text{Conversion rate (\%)} = \text{number of converted visitors} / \text{total visitors} \times 100$$

For E-commerce stores the conversion rate ranges between one and five percent (MarketingSherpa.com, 2014).

According to Croll, the Conversion rate is considered to be the most important metrics in the e-commerce field. (Croll, 2009)

In other words, this metrics shows how visitors of the website can become buyers, the ones who contribute to business goal, and a user-oriented website can very well contribute to reaching it.

Since different companies may have their own methods of calculation Conversion rate, it is worth mentioning the one of them that is very well-known in e-commerce world – Click-Through Rate (CTR).

Click-Through Rate

CTR (Click Through Rate) - the ratio of the number of clicks on a link to the number of times in was displayed. This metric helps determine the effectiveness of a particular platform for advertising, the effectiveness of the advertising campaign as a whole, as well as individual advertisements (Google, n.d.).

$$\text{Click-through rate} = \text{number of clicks on the link} / \text{number of times displayed} * 100\%$$

Average order value

Average order value (AOV) shows the solvency ratio of the audience and illustrates the ratio of the value of all completed orders to their quantity. With it, you can predict the store's revenue (knowing the traffic to the site and the conversion to purchase), and also edit the marketing strategy in time (Optimizely, n.d.)

$$\text{AOV (\%)} = \text{completed orders value} / \text{orders quantity} \times 100$$

Shopping cart abandonment rate

This coefficient reflects the number of visitors who added goods to the basket but did not complete the order. According to a SaleCycle study, 76.9% of users leave the cart page and do not return for purchases. (Charlton, 2018)

$$\text{Cart Abandonment Rate (\%)} = 1 - (\text{Number of Completed Transactions}) \div (\text{Number of Shopping Carts Created}) \times 100$$

H.E.A.R.T measurement model

The HEART framework (Happiness, Engagement, Adoption, Retention and Task Success) was created at Google and is used most often in the digital sphere. It can also be applied to evaluate the quality of UX on the website. (Rodden, 2010)

Table 2 HEART framework (Source: Rodden, 2010)

Framework element	Description
Happiness	Refers to a user's subjective attitude towards the product. Examples of metrics are i.e. user satisfaction, ease of use of the website.
Engagement	Level of interest in the product. Starting to use the product, the user continues to do so. Examples of metrics are the frequency of user visits, depth of interaction.
Adoption	Used for counting the number of unique visitors of the website at a given period of time.
Retention	Tracks how many becomes loyal to the website and returns to it. "Loyal" term definition in this case can vary depending on the product or business—examples of metrics: the number of repeat purchases, user activity during period of time, etc.
Task Success	The success of key tasks. These are technical indicators such as efficiency, productivity, and error rate. Examples of metrics can be the speed of search for information on the website.

UX target table

According to Hartson and Pyla, it is difficult to determine whether the design of a product (in our case – website) in meeting UX goals (Hartson and Pyla, 2012).

They proposed a simple concept of the UX target table that is showed below.

The first column should contain along with related user class, which is based on the current experience of a user or limitations. UX goal usually describes the desired user experience, i.e., fast checkout on the website. UX measures can be divided into two parts: objective (can be measured, i.e., first performance) and subjective (based on the user's opinion).

Measuring instruments are typically benchmark tasks (BT), i.e., to purchase an item or questionnaires. UX metrics represent what needs to be measured, i.e., checkout time. A baseline can be an average time of numerous performed tasks. The target level is a result

that is hoped to be achieved, and respectively, the observed result is the result that is accurate now.

Table 3 UX target table – e-commerce website example (Source: author's own processing according to Hartson and Pyla, 2012)

Work Role: User Class	UX Goal	UX Measure	Measuring Instrument	UX Metrics	Baseline Level	Target Level	Observed Results
Potential buyer: new user	Low error rates for new user	Initial user performance	BT1: checkout with two cart items.	Avarage time for a task	3,5 sec	1 sec	4,5 sec
Potential buyer: experienced user	Fast loading	Initial user performance	BT2: checkout using a promo code	The average number of errors	<1	<1	2

3.4 Why do some e-commerce businesses fail?

Comparable to any other business, e-commerce businesses might also struggle to succeed in the long run. According to multiple sources online, the failure rate fluctuates from 80% to 98% these days (Labrien, 2016).

Given so many benefits of e-commerce, such as no barriers to entry, low overhead costs, and flexibility (Mulačová, 2013), some companies find it hard to thrive in the industry. There are dozens of reasons why an e-business might fail, and each business has its circumstances. The author analyzed various articles on the topic online to identify the most common reasons in the table below.

For the purposes of this thesis, it is important to gain an understanding of why an e-commerce website can fail and what is the reason behind it.

Table 4 Common reasons for e-commerce failure (Source: ReadWrite.com (2019), Appliedi.net (2016), Outerboxdesign.com (n.d.), Ecommercewarriors.net (n.d.), Sinnot (2017) and Neil (2019)).

Reason	Description
Wrong product	This is considered to be a crucial aspect of an e-commerce business. Often it might be products that do not have a market need yet or the outdated ones.
Lack of proper marketing strategy	An appropriate marketing strategy is something every e-commerce business needs to have in order to maintain i.e., a high rank in Google search results. It is essential to market the e-shop and its products among different communication channels.
Competition	Competition is a huge part of any business, especially online. It might be a problem for new online stores that want to compete with big and known platforms or businesses.
Website design issues	This may include lousy category navigation, a bad layout, website not being responsive, absence of critical UI elements, or the time it takes for pages to load.
Inappropriate (wrong) pricing of the products	It is essential to set reasonable pricing, and it is very specific to the type of products the business is selling online.

The identified common reasons for e-commerce website failures are selling the wrong products, a lack of marketing strategy, underestimated competition in the industry, website design issues, and inappropriate pricing.

For further discussion in this thesis, the author chose one of the common reasons, which was Website design issues.

According to Salmon (2008), one of the ways to recognize a quality e-shop is to make sure that it has a quality design. “The customer can go elsewhere with a single click, so more than any other website, and e-shops must ensure that they perform well and navigate well. The website should be user-friendly and well-arranged, with an emphasis on easy access to all important information.”

McKown (2017) stated that lousy website design is a big red flag when purchasing online.

Hong Kong Retail Management Association (n.d.) also highlighted the same aspect in their Quality E-shop Recognition Scheme and pointed out the importance of user-friendliness of the website as one of the critical points of quality.

Baymard Institute, an independent web usability research institute, conducted various studies and defined the average cart abandonment rate, which sits at around 69% with some of the listed reasons for abandonment being long checkout processing times (21%) and website errors (13%). (Baymard Institute, n.d)

4 Business value measurement of UX/UI design in E-commerce

In order to understand the value UX/UI design can bring to a business, there have to be a way to measure results that UX/UI design changes cause.

In the previous chapter there were discussed some of the possible ways to measure the impact of current state of an e-commerce business, however, they might very well be applicable to measuring the changes that occur from start to finish of the project.

4.1 Return On Investments (ROI)

According to Jack and Patricia Phillips, Return On Investments is one of the basic indicators of the profitability of investments and the success of a campaign or website in general. It can be applied to the entire campaign, channel and message (Phillips P. and Phillips J., 2008). ROI can be calculated the following way:

$$ROI (\%) = (Net Profit / Investment) \times 100$$

It is worth mentioning that ROI should not be considered as a financial projection and should only be treated as a financial exercise, thus it is an estimated UX metrics. Design improvements are cumulative over a period of time, and it is worth to pay attention to ROI long-term, because in the first year of implementing improvements the results might not be informative (Moran, 2020)

ROI might also be questioned, because it can be applicable to any activity that took place regarding e-commerce.

4.2 UX Balanced Scorecard (BSC)

The Balanced Scorecard (BSC) is one of the most comprehensive models of a strategic performance measurement system. The method was defined by Robert S. Kaplan and David P. Norton. As part of their practice in companies, they understood the limited ability of mere financial indicators to determine, measure and control the success of the implementation of a corporate strategy. They also recognized the need to develop a comprehensive model that would allow real objective measurements. (Kaplan and Norton, 1996).

Turner described the well-known Balanced Scorecard system that connects UX/UI activities with company KPIs.

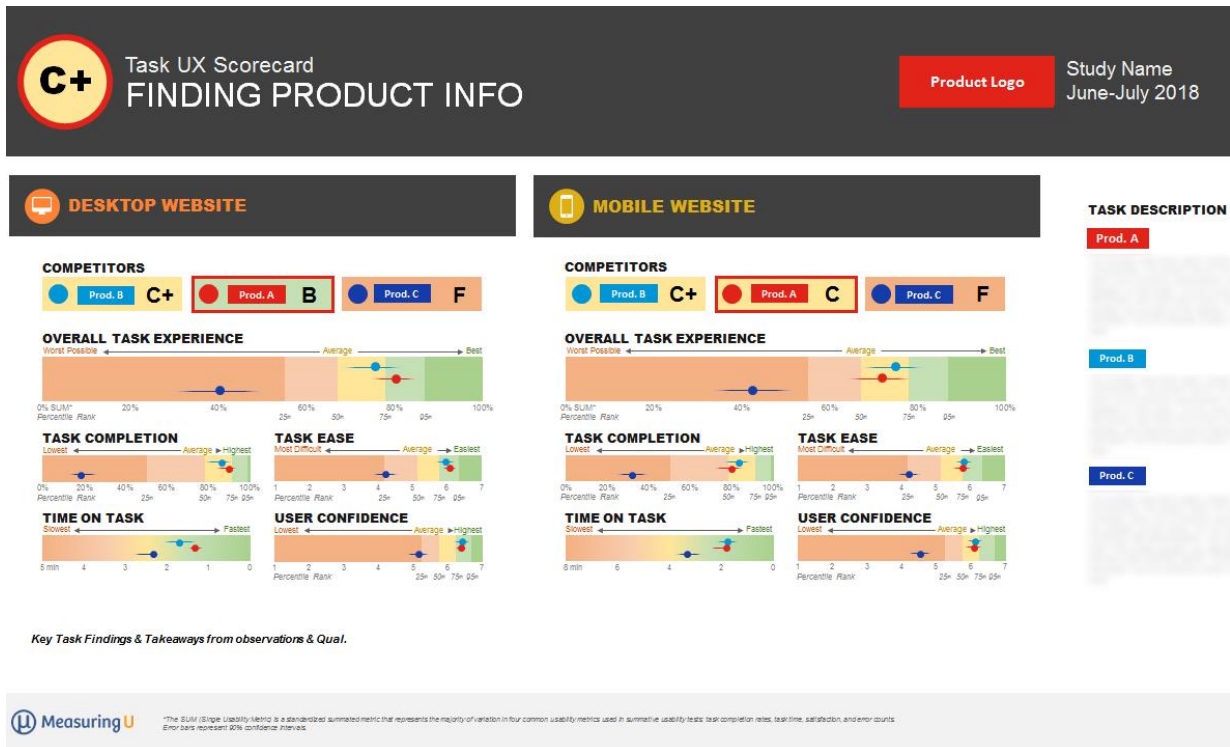
The model is described using the following measures (Turner, 2011):

- **Financial Perspective** is about providing timely and precise data on capital, processing, and support of financial data. It usually involves measures such as revenue, costs, ROI, etc.
- **Customer Perspective** implies customer orientation and customer satisfaction in any field. Even if the current financial situation is quite prosperous, it is necessary to continually analyze the types of customers, unique visitors, processes, products, or services, their compliance with the expectations of customers to ensure customer loyalty in the future.
- **Internal Business Perspective** - These indicators allow us to determine the customer focus of the company. Such work cannot be assigned to external specialists, because here we need a clear view of all the organization's business processes.
- **Innovation and Learning Perspective** - Employee training promotes corporate culture. In any company, educated people are the leading resource. Sufficient UX skills are required most of the time when talking about customer satisfaction and internal processes.

PERSPECTIVE	USER EXPERIENCE STRATEGY	
FINANCIAL	Increase revenue	Reduce costs
CUSTOMER	Increase conversions, retention, and loyalty	Increased effectiveness (task completion) and efficiency (time on tasks) and reduce Training and Support
PROCESS	Conduct user research and iterative design review with your customers	Conduct usability evaluations with your customers' end-users
EMPLOYEE	User Researcher Information Architect Visual Designer Interaction Designer Usability Engineer	

Figure 19: BSC – User Experience processes (Source: Product Camp SoCal, 2010)

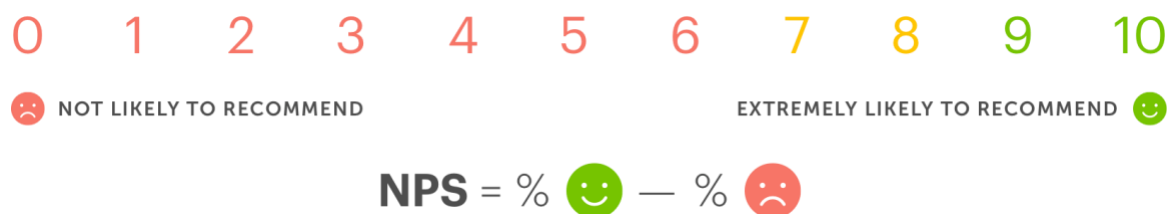
There can be found some other variations of UX BSC, i.e. one model presented by MeasuringU that can be directly applied to E-commerce that include mobile and desktop versions.



4.3 Net Promoter Score (NPS)

Net Promoter Score (NPS) a metric specifically designed to determine the level of customer loyalty to a product, brand or company, as well as its willingness to recommend your product or service to its environment.

For the first time, the concept of such an assessment of loyalty was proposed in 2003 by Fred Reicheld, who was the first to think about the lack of a tool for measuring loyalty to companies from different industries or measuring this indicator among customers of different companies working in the same niche (Reichheld, 2003).



The clients participating in the survey are divided into three conditional groups:

- Promoters (9 to 10)
- Neutrals (7 to 8)

- Detractors (0 to 6)

After receiving the necessary data, the loyalty index is calculated by subtracting the percentage of detractors from the percentage of promoters.

Even though the model receives criticism, it is still being used by the companies from Fortune 500 (Customer Guru, n.d.)

5 Research of UX and UI business value and influence

In this chapter will be focused on the practical part of this thesis, which is dedicated to the research of the business value of UX and UI principles, as well as its influence on the most common performance measurement indicators. It is achieved by the qualitative research methodology that includes surveys and expert interviews.

Individual subchapters thoroughly describe methodologies that were chosen and the results that were achieved during the research.

5.1 Methodology - surveys

The research was partially carried out using anonymous electronic questionnaires. This method was chosen primarily because of the possibility of rapid dissemination among its respondents and, at the same time, the rapid response rate in terms of time.

For a better understanding of the problematics of this thesis and to find out since the whole topic of User Experience is about user-centricity, the first survey concentrates on the understanding of consumers and people that have experience with online purchasing of goods and services. That way, it is easier to make conclusions on participants' attitudes towards the topic.

The second survey was shared among the experts and professionals in e-commerce and User Experience areas to get their opinion on the researched topic.

The questions and statements in both questionnaires were picked based on the theoretical discussion in the previous chapters and were composed using the Likert scale method, which is the most common approach to scaling answers in a survey study

The Likert scale for this research was a 5-digit relationship scale, in the application of which respondents had to determine the degree of agreement or disagreement for each set of statements about the objects under consideration.

The Consumer Survey was based on the Agreement scale:

- 5 - "Strongly agree",
- 4 - "Agree",
- 3 - "Neither agree nor disagree",
- 2 - "Disagree",
- 1 - "Strongly disagree".

The Expert Survey was based on the Agreement and the Importance scale that was structured as follows:

- 5 - “Extremely important”,
- 4 - “Important”,
- 3 - “Neutral”,
- 2 - “Low importance”,
- 1 - “Not at all important”.

1.1.1 Consumer Survey

A total of 189 respondents participated in the questionnaire. The targeted audience was people that have any kind of experience with online purchases, whether it be services or the actual physical products.

The first section of the questionnaire was focused on basic data.

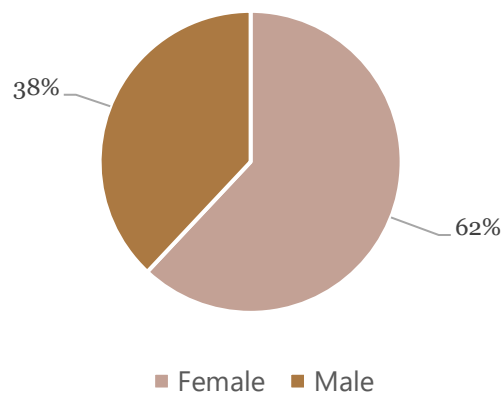


Figure 22: Gender of the respondents (Source: author's survey)

The age range was divided into five categories. The most numerous of them are 18-24 years old, with 90 respondents. The following categories are 25-34 years with 76 respondents, 35-44 years with 15 respondents, 45-54 years with four respondents, 55-64 with one respondent, and 65+ with 0 respondents.

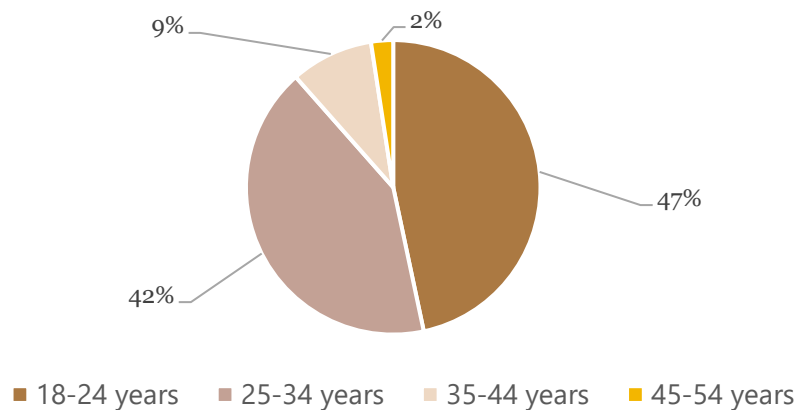


Figure 23: Age groups of the respondents (Source: author's survey)

To make sure the participants have experience internet shopping, they were asked to specify how many times per year do they purchase services or products online roughly. 59,3% of respondents confirmed that they make purchases online more than ten times a year. 25,4% limited their purchases from 5 to 10 times a year. 12,4% selected the option from two to five times a year. The rest 2,8% confirmed that they make one or no purchases online in a year.

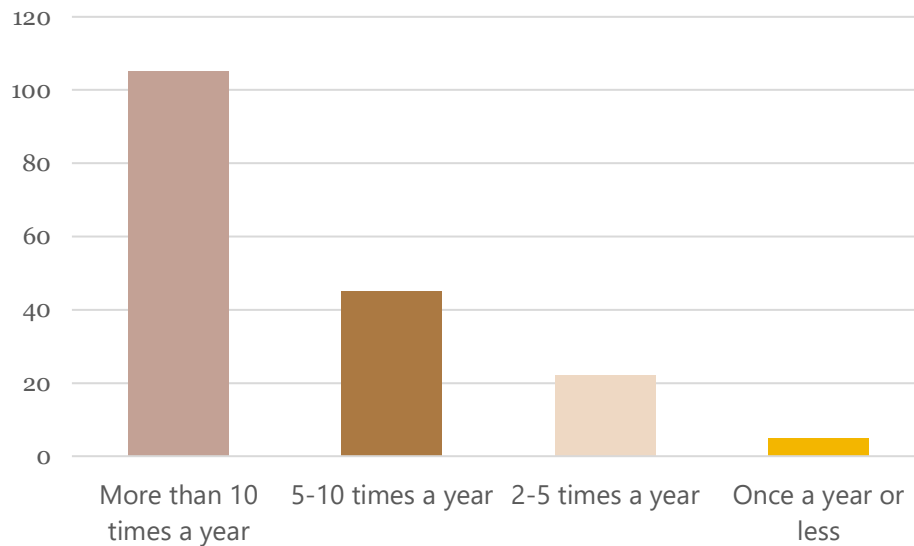


Figure 24: Online purchasing frequency of the respondents (Source: author's survey)

In the following section of the survey, the respondents had to agree or disagree with the statements, most of which were based on the UX principles described by Morville and Usability Heuristics described by Nielsen in Chapter 2.

The statements and the principles that each of them are linked with were:

Table 5 Statements and principles – Consumer Survey (Source: author's survey)

Statement	Referred to principle
I am less likely to purchase on the website if it is unavailable in the language I understand.	Accessibility (Morville)
I am less likely to purchase on the website if the pages take a long time to load	Flexibility and efficiency (Nielsen) and Desirability (Morville)
I am less likely to purchase on the website if it does not work well on my mobile phone	Accessibility (Morville)
I often judge the website credibility and trustworthiness by how it looks	Desirability (Morville)

I typically do not read large paragraphs of text on the website	Aesthetics and minimalism (Nielsen)
I experience negative emotions when I cannot complete my purchase easily	Error prevention (Nielsen)
I think online-shopping websites should be useful and easy-to-use	
I am less likely to return to the website if I had a negative experience with it	

Statement #1 and Statement #3 refer to the **Accessibility** principle that explains the need to have the website be accessible by all people, that includes users that might find the need to use their phone to make purchases or people that might not speak in the language the website is localized to. Statement #2 describes both **Flexibility and efficiency** and **Desirability** principles that are responsible for the functional appeal of the website. Statement #4 also describes the **Desirability** aspect of the website, but in this case, its visual appeal. Statement #5 is referring to the **Aesthetics and minimalism** principle, explaining the need to only include the necessary information without clutter. Statement #6 referred to the **Error prevention** principle by Nielsen.

In this section of the survey, respondents had to agree or disagree with the statements mentioned above. Out of the first four statements, it is easy to determine the mean values of individual statements., given that every answer has its own numeric value.

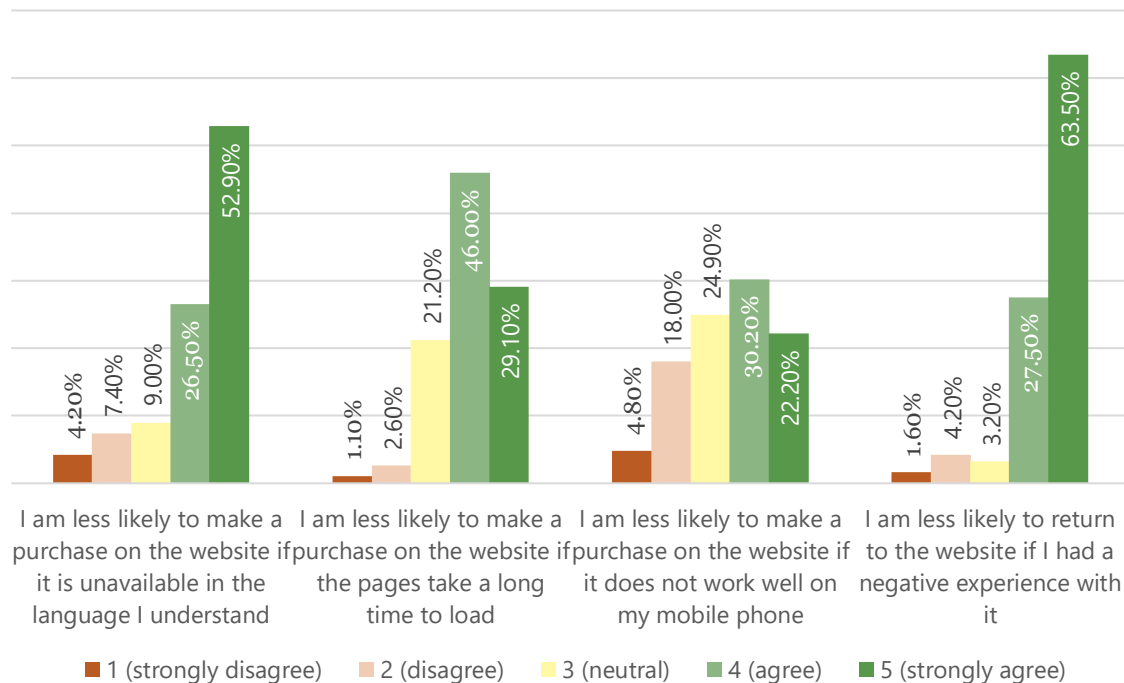


Figure 25: Individual statement's responses — set 1 (Source: author's survey)

As shown in Figure 5.4, statement #1 (from now on S) turned out to be the one with the highest mean value of 4,2 and has the highest amount of 5 (strongly agree) votes. Following with S2 with 4 mean value, S4 with 3,9 and S3 with 3,5.

According to the results, S1 received close to 80% agreement rate, which suggests that people are a lot less likely to purchase on the website is unavailable in their language.

S2 responses suggest that 75% of respondents might not have purchased on the website that has a slow loading performance. 25%, however, would not have a problem with it and proceed with the purchase.

S3 results turned out to be ambiguous, with them being close to 50/50 split and will need further explanation in the results summary. S4 suggests that 72% of respondents might, at some point, judge how credible and reputable the website by its cover.

The next set of statements shown in Figure 5.5 are based on the subjective opinions of the respondents but can provide the idea of people's attitudes toward websites' performance and usability.

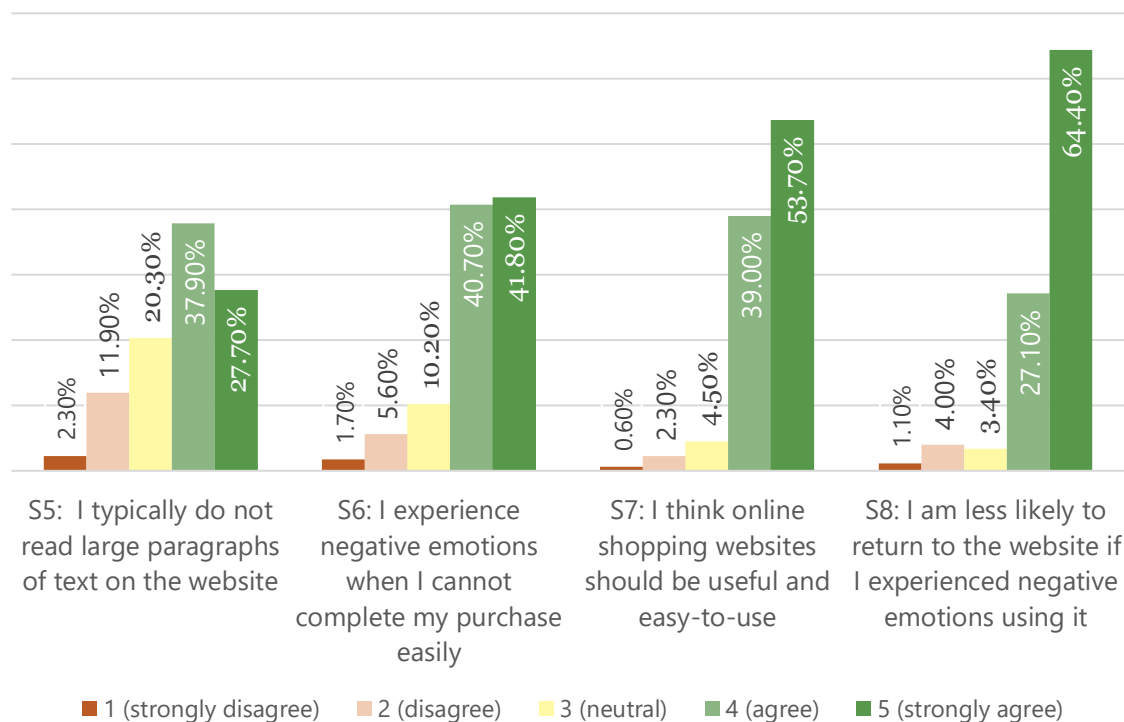


Figure 26: Individual statement's responses — set 2 (Source: author's processing)

S8 received the highest amount of "strongly agree" responses and tells that more than 90% to some extent agree than they might not return to the website if they had a negative experience with it. Along with S6, which has a 'negative emotions' part in it, it can be

concluded that the complicated process that involves order checkout will most likely result in the potential customer not returning to this website anymore.

More than 65 percent agreed to some extent that they might not always read large paragraphs of text on the website in the S5, which was also analyzed by several studies dedicated to using an eye-tracking technique to find the most common areas where users focus their attention on while browsing the websites.

1.1.2 Expert survey

The survey that is described in this subchapter was conducted among the professionals that have two and more years of experience working in the areas of E-commerce and UX/UI design. The goal of this survey was to get to know the opinions of experienced people in the E-commerce driven business area.

The experts were reached out to using a popular social media platform LinkedIn that makes it easily find a large number of people with working experience in different professional areas. All the respondents were individually picked out, and the total number of those who participated is 26.

In the first section of the questions, respondents had to answer the question “How important are the following to the success of an E—commerce business?” by rating the importance of individual statements according to the Likert Scale described previously. Statements in this section include ones that were based on Morville UX honeycomb describes in Chapter 2 as well as those that are not related to UX or UI areas, but also essential to mention.

Table 6 Statements and principles – Expert survey (Source: author’s survey)

Statement	Referred principle
S1: The user is confident that the e-commerce website is secure.	Credibility
S2: The e-commerce website offers useful information and features for evaluating products.	Usefulness
S3: Various payment methods are available.	Nonrelated to UX or UI areas
S4: All users of the website have the same experience, no matter what device they are using, the language they speak, or any disability they may have.	Accessibility
S5: The products are priced appropriately for the market	Nonrelated to UX or UI areas

S6: The visual design and branding of the e-commerce website help to make its products more desirable than their competitors.	Desirability
S7: The e-commerce website delivers a valuable experience to the user.	Valuableness
S8: The user can find products that are relevant to them.	Findability
S9: The user finds the website intuitive and easy to use.	Usable

The second section of the survey includes the statements that were discovered during the analysis of credible sources on UX and UI as well as non-credible. The results found will be provided in a list of most-common and most-discussed benefits that can be applied to E-commerce and were further included in the survey to let the experts rate their legitimacy from their experience.

The statements found from the review are not yet considered to be accurate since a portion of the sources are not academic studies and do not have precise data to support the claims. Therefore, each of them should be considered unconfirmed or potential.

All the results listed in the comments below were picked out according to the sources listed in Annex A.

From the review of the articles, there were identified seven potential points of influence that can contribute to the e-commerce based business:

1. Good UX/UI website design promotes increased sales and revenue for the online business.
2. Good UX/UI website design can reduce support costs in the long-run.
3. Good UX/UI website design reduces development costs and time in the long-run.
4. Good UX/UI website design contributes to increased customer productivity when purchasing online.
5. Good UX/UI website design provides a competitive advantage.
6. Good UX/UI website design increases customer satisfaction.
7. Good UX/UI website design contributes to increased brand loyalty.

Each respondent had to agree or disagree with each of the given statements according to the Likert Scale described previously.

Results – First section

The first question on the survey was asked to figure out the experience that individual respondents have with E-commerce, and the results can be seen in Figure 5.6.

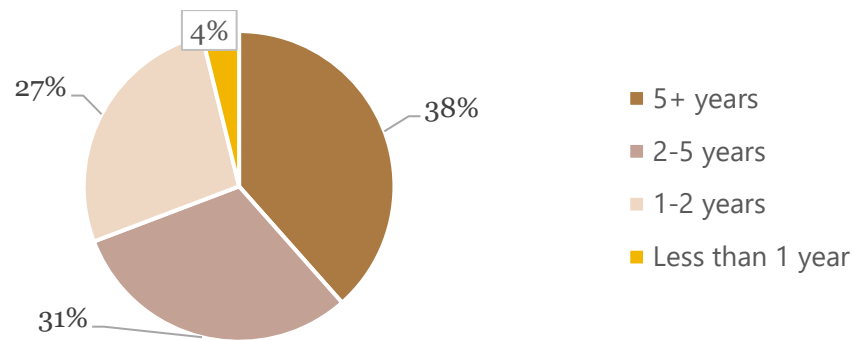


Figure 27: Experience working in E-commerce (Source: author's survey)

The results from the first set of statements from S1 to S5 demonstrated in Figure 5.7 show that the majority of respondents, to some degree, consider most of the statements mentioned to be necessary.

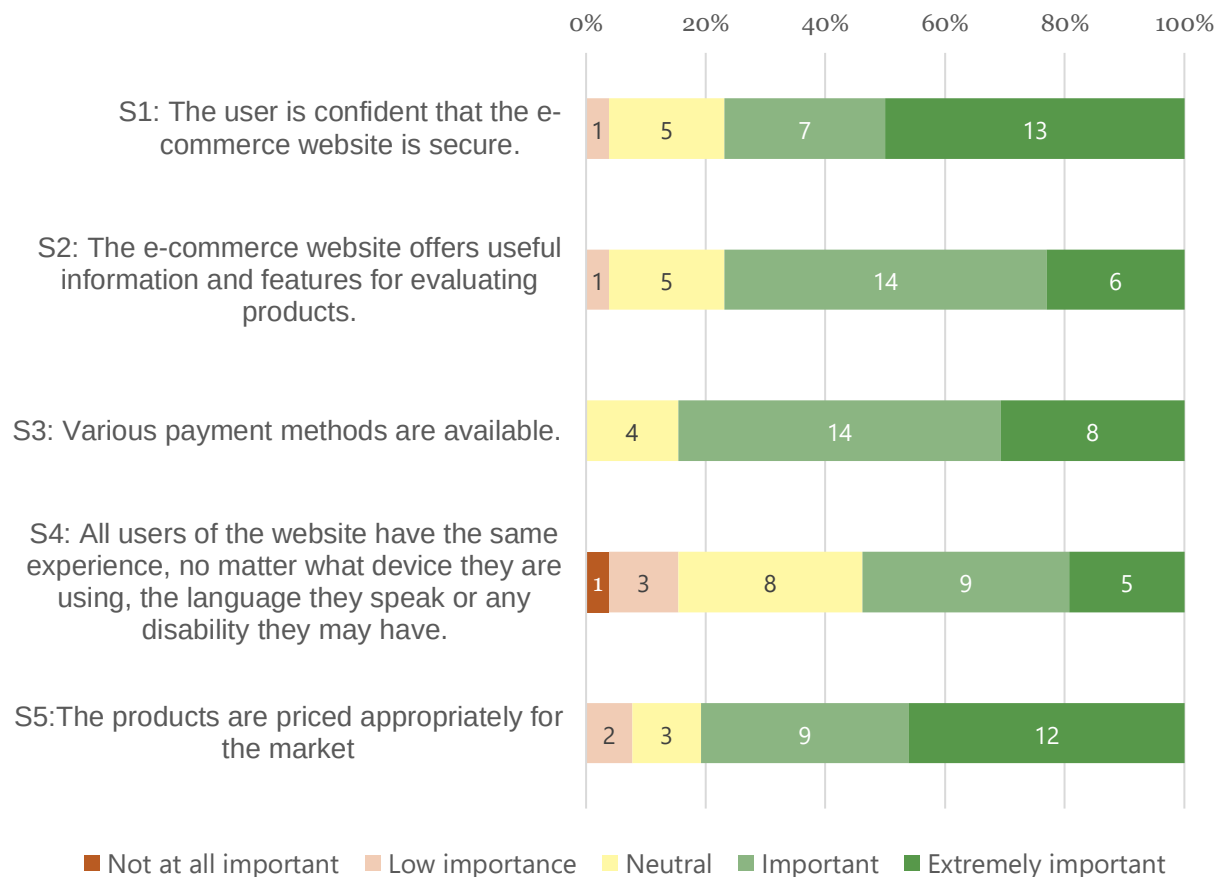


Figure 28: Individual statement's expert responses — set 1 (Source: author's survey)

Among the statements that were directly related to UX and UI principles, respondents consider being important the point that the user needs to feel confident that the e-commerce website is secure (S1). Therefore, it is essential to make him feel this way by letting him know. The usefulness of information on the website (S2) was also considered to have a high value of importance from the respondents' point of view, with 77% above midpoint value.

S4 received mixed answers, and one of the respondents specified in his comments to the survey that “no matter how hard we try, a blind person will hardly have the same experience as a person without such a disability” and suggests that it is tough to achieve the same experience for all people.

S3 and S5 are the statements without any relation to UX or UI received the highest ratings among the entire set of statements, which also suggests the importance of those points and that they should be overlooked.

The next set of statements (S6-S9) shown in Figure 5.8 received a meager amount of below midpoint answers, and overall most of the respondents chose ‘Important’ or ‘Very important’ option to each of the statements.

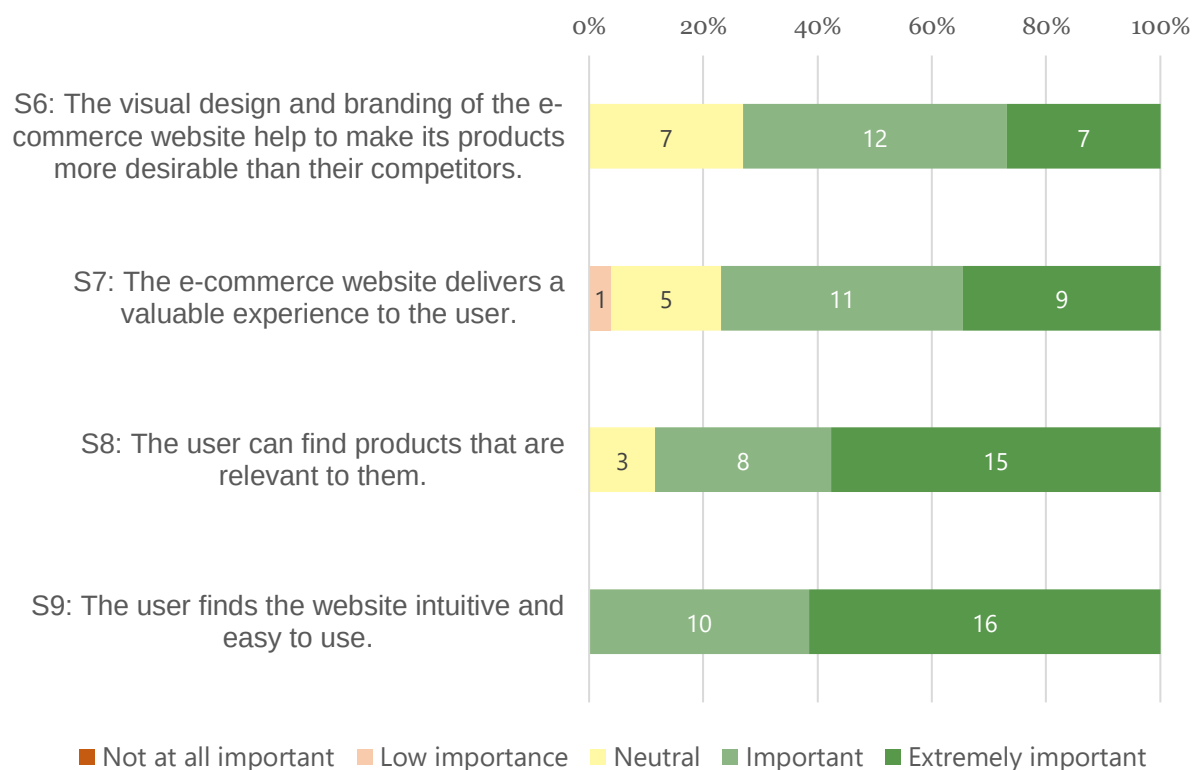


Figure 29: Individual statement’s expert responses — set 2 (Source: author’s survey)

Respondents 100% agreed that it is crucial that the website needs to be intuitive and easy to use from the user’s perspective (S9), and at least 90% of respondents confirmed that the user should be able to find the relevant products (S8). Therefore, it is necessary to reduce any friction when it comes to this process.

Results – Second section

In the further section of the survey, respondents were asked to agree or disagree with the statements that were based on the review of a number of online sources.

The third set is shown in Figure X. The claim that UX and UI design can promote increased sales and revenue was highly agreed to by the respondents, with only 3 of them voting for a 'Not agree/Nor disagree' option.

Statements that claimed that UX and UI design could reduce support costs and reduce development time received mixed answers. The second received 6 disagreeing and 11 neutral responses, which suggests that investing can increase development cost in the long—run or do not affect it in some cases.

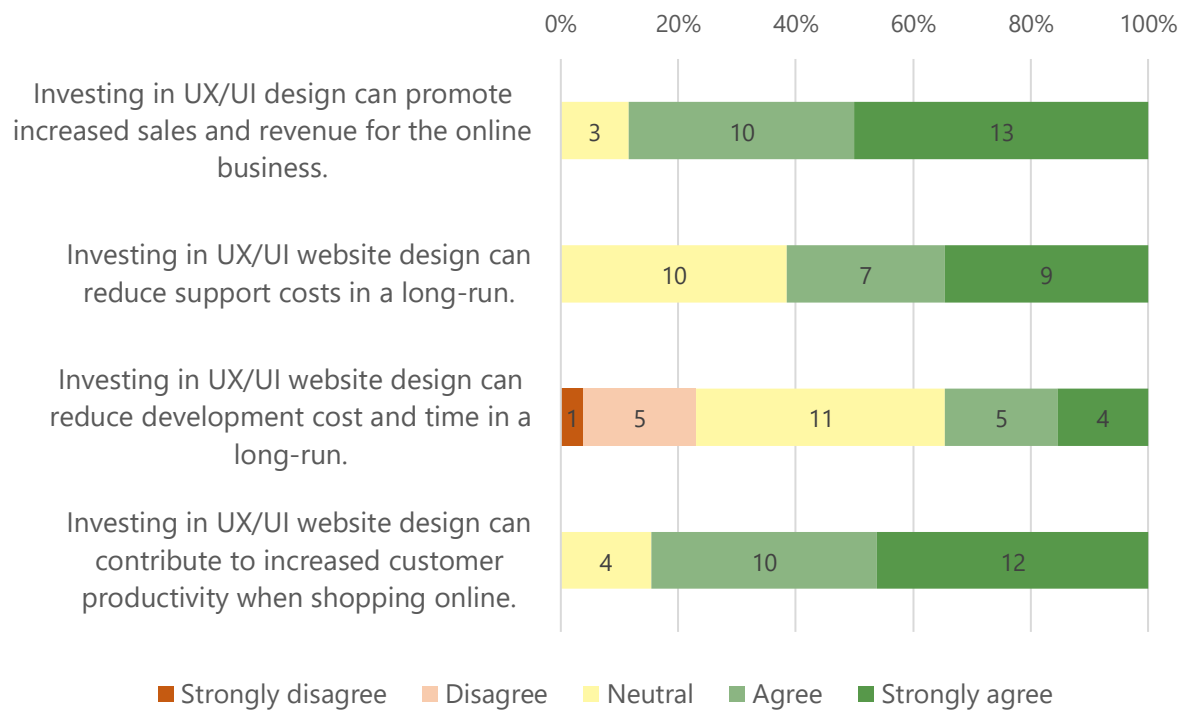


Figure 30: Individual statement's expert responses — set 3 (Source: author's processing)

The fourth set of statements received mostly above midpoint responses. The first two statements receive a large amount of 'strongly agree' responses, and that suggests that the design might, in fact, provide a competitive advantage to the business and increase customer satisfaction, which is what the whole idea of UX is about. The last statement which stated that UX/UI could increase brand loyalty was mostly agreed to, however, still received two disagreeing responses.

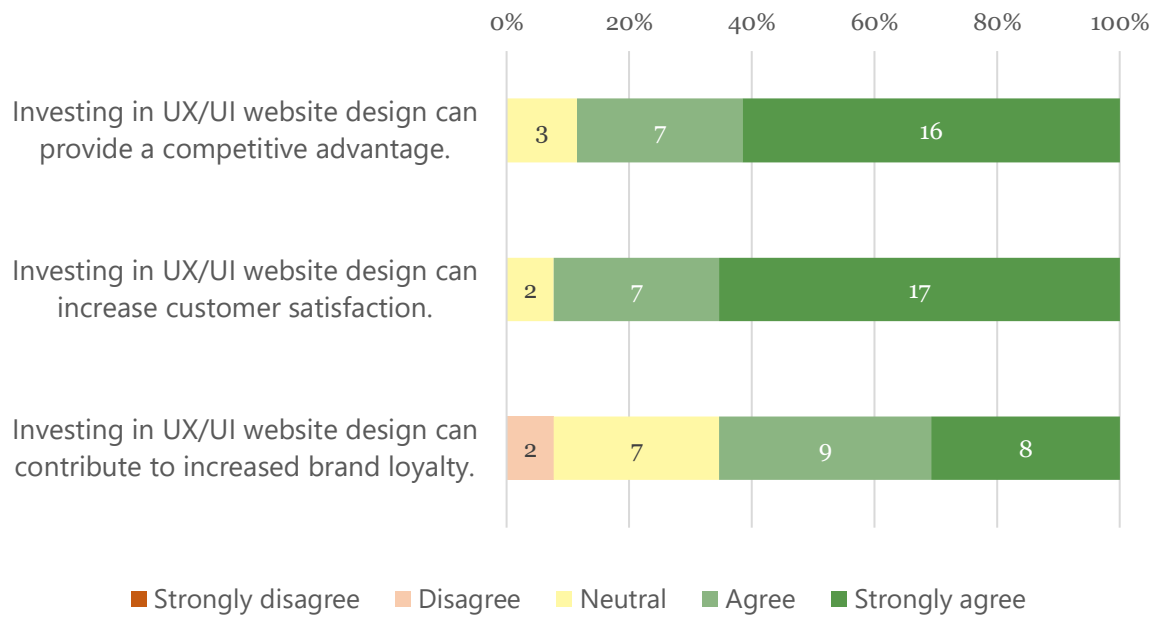


Figure 31: Individual statement's expert responses — set 4 (Source: author's processing)

5.2 Methodology - interviews

For further data collection, it was decided to have semi-structured interviews with the experts in E-commerce. The conducted interviews were based on verbal communication with one person. The questions that form the framework of the whole interview were prepared in advance. These questions are based on the facts found above in this work and the goal of the thesis. However, with semi-structured interviews, there is also the possibility to deviate from the questions and ask questions that the interviewer considers important for this area or further development of the interview. This way, it was possible to get very authentic and important opinions for this paper.

The topic of the survey and the interview was explained to the respondents as a research of role of UX and UI design in e-commerce, any hypotheses of the researcher were not shared with the participants to avoid bias.

The interview took place in the time period of April 2020, and three respondents were interviewed:

Table 7 Statements and principles – Expert survey (Source: author's survey)

Participant	Name of the participant	Position	Interview type	Length
R1	Vojtěch Stříbrský	Online Sales Manager (website e-commerce)	Remote via Zoom online conference tool	30 min

R2	Roman Christov	Senior E-commerce Specialist (desktop e-commerce)	Remote via Zoom online conference tool	30 min
R3	Petr Chlup	Senior E-commerce Specialist (mobile e-commerce)	Remote via Zoom online conference tool	30 min

The goal of the chosen research was to get professional opinion towards the role of UX and UI in E-commerce. The expert who participated in the interview are tightly involved with E-commerce, with some of them are working on different e-commerce platforms, those include Desktop applications and Mobile applications (Android and iOS). The individual transcripts of each discussion are provided in Annex B.

When compiling the research questions, the author tried to use different types of questions starting from the easier ones and moving to complex ones. The IPA method will be used to evaluate the interview. That helps with the in-depth examination of how a person shapes the meaning of his experience, which allows understanding an individual event or process - a phenomenon (Smith and Flowers, 2009).

1. Can you please introduce yourself and what you do in Avast?
2. How closely does your teamwork with UX and UI designers? Describe the process?
3. Have you ever taken part in the redesign or optimization projects on the website?
4. What are typically the business objectives and KPIs of these projects? How do you measure whether it was successful or not?
5. As an E-commerce expert, you can see the effect of any little changes on the website/app. Can you think of an example of the design change that has been made, and what was the result of this change?
6. Do you think that UX/UI design creates value for e-commerce business? Why?

Interviews summary

This part of the chapter provides the IPA analysis of the interviews that were conducted for the purposes of this research.

Table 8 IPA analysis - interviews (Source: author's processing)

Questions	Answers
How closely does your teamwork with UX and UI designers? Describe the process.	R1: Design is essential and we work very closely with all of the designers. Those are visual designers and UX designers. And I think the visual design is essential and UX is also crucial for customer understanding. We have a very tight cooperation with the design team and we discuss every single button on a page in order to meet the sales

	expectations. And to make sure it resonates with the customer. R2: We work with them quite a lot. All the content we are displaying in this matter within the app, all of it need to go through UX designers, UI designers and copywriters before it reaches out developers. We are trying to optimize all of the sales flows constantly to provide better performance. R3: Very closely. Basically, for us, it's essential to create any sales activity. So basically, we go through the circle that includes copywriters, the designers, both UX and UI. It always depends on what kind of activity it is. Every single activity that is being done, we make sure to always include the design team.
Comment: All the respondents confirmed that they work very closely with UX and UI designers and making sure that they are in tune with each other to make the best possible decisions when it comes to sales funnels and profitability of the company.	
Have you ever taken part in the redesign or optimization projects on the website?	R1: Sure. Redesign of the web usually relates to a bigger change either in a product or in a branding. So basically, we did both. We wanted to be a little more product-oriented in terms of communication. R2: Of course. We do them quite often. R3: Yes. We are forced to constantly innovate and try to bring new things to the table. So a part of our job is constantly testing. It can be the Home screen, an overlay, it can be notifications.
Comment: All three respondents admit that they took part in a redesign or optimization projects. Petr pointed out that most of the time, they are forced to provide refreshments and updates quite regularly, and Vojtech explained that the updates are typically made in order to better the communication with the customer.	

What are typically the business objectives and KPIs of these projects? How do you measure whether it was successful or not?	R1: For monetization part, it is always the Conversion Rate. So basically, we want to make sure that your conversion doesn't drop at least and we keep an eye on how users are converting and if there is any upside. When measuring success, it usually really depends on what page we are talking about. But generally, we typically measure RPM, which is Revenue per Mille and AOV (Average Order Value). R2: If the changes were already made, we might look at the number on clicks change and the RPM measure to tell whether the change was successful or not. R3: We look at RPM measure, which is Revenue per thousand people. If we make any changes to the CTA (Call-To-Action) button and the RMP metrics value increases than we know that we're on the right track. We also have RPI metrics, which is revenue per install, but this one is very app-specific. Generally, there are thousands of different metrics that we look on including Conversion Rate, the Average Order Value and so on.
Comment: All of the three respondents mentioned RPM (Revenue per Mille) metric that helps them with measuring the success of business objectives that were set before the change. Vojtech and Petr both pointed out the Average Order Value metrics and how they use it. Aside from RPM, Roman also mentioned a number of clicks for various buttons.	
As an E-commerce expert, you can see the effect of any little changes on the website/app. Can you think of an example of the design change that has been made, and what was the result of this change?	R1: We completely changed the layout of our pricing page recently. It had a lot of unnecessary things in it that we removed and we also wanted to highlight the purchasing plan that is the best for us in terms of profitability, which is a 36 months. So yes, we did the UX and layout optimisation on this page that really improved the Conversion Rate. On the Home Page of our website we have to make sure that we communicate the money-back guarantee so that we can be

	trusted and that alone has raised the conversion rate of this button up to another 5%. R2: This is a tricky question. From my professional experience, I've seen the changes in Conversion rate uplift ranging from 5%, which to me is a flat result, up to 20%. We've seen a lot of effects after changing UX and UI of the app, positive and negative. The only that I constantly talk about is that designers are not very sales oriented people. It is very important to analyze the data to make sure that you make correct decisions. R3: From my experience, I've seen very little and subtle changes that had a huge impact. We are always trying to emphasise what the deal for the user that will bring us the biggest profit. Even if it's a small change in the way we communicate our prices make a huge difference. I can't share the exact numbers and data but it really makes quite an impact on multiple metrics including Conversion Rate.
Comment: All three of the respondents agreed that they have seen shift of various indicators after the changes in design and pointed out the improvement in Conversion Rate starting from 5%. Vojtech explained the adjustment on the Home Page and how a very small change in the information alone can raise the Conversion Rate. Petr also noted that they have seen subtle changes being made that made a big difference in the indicators they usually look at.	
Do you think that UX/UI design creates value for e-commerce business? Why?	R1: We improve the look and feel, we get increased sales and performance, we communicate and present our products in a better way. But it is very important that both design and Sales teams to understand each other. R2: So it for sure does [bring value], but as I said, it is good to rely on data. You have to be able to read numbers in order to do correct decisions. I also want to say that it

	doesn't bring the never-ending value, if that makes sense. Even the team with the best designers in the world can't produce constant growth as the exact pace. If we were to wake up every day with some groundbreaking ideas we would probably be multi-billionaires by now. From my experience, one of the most important things when it comes to conversion of our users is whether these users engage with the app, so in this sense paying attention to UX is completely quintessential for us to be profitable, which is similar to the web but from my experience it is much more important on web. R3: E-commerce is a very dynamic environment, you can leave the app or the website in a matter of half a second and you're gone. And you never even remember where you were a second ago. The duo of UX and UI is what ultimately creates a good customer experience, which supports sales. I don't think that we would be able to reach our company goals and objectives without investing into these two areas.
Comment: All three respondents that in order to make the best decisions when it comes to design changes and optimisation all the teams involved need to communicate as good as possible. Respondents were all talking about the same issue between Design and Sales teams and how essential it is for them to cooperate better and make compromises on different things. Since designers are most of the time very consumer oriented people it is very easy for them to get stuck in the area where it is all about customer satisfaction, they need to present their work in a form that with also bring monetary value for the company. Roman made an interesting point which is that it is impossible for UX and UI design to bring a never-ending value and even the team of best designers won't be able to achieve the constant and ongoing growth.	

6 Discussion

This chapter is dedicated to summarise the findings of the practical part of this thesis and answer the questions „What are the point that can be influenced by UX and UI principles in E-commerce?“ and the additional question “Is it worth for a business to invest in UX/UI design?”. The chapter will as well address the limitations that were discovered and provide recommendations for the further research on the topic.

6.1 Interpretation of the results

Any visitor or a user of the product should be considered valuable because this is ultimately the group of people that drives the revenue of the company or business.

The results from the **Consumer survey** show that even without knowing, the visitors and users are aware of the design. The audience was made up mostly of seasoned online shoppers, with 85% purchasing five or more times in a year. The research has shown that more than 70% of respondents might, at some point, judge how credible and reputable the website by its cover, and it is a high number to dismiss.

Suitability on a mobile phone was much lower than expected. It compares differently to the survey "Whitepaper: Web-Design and Business Success" (Brackett, 2014), where 76% of respondents, to some extent, agreed that it is crucial for them if the website is not responsive (designed for mobile platforms). However, it may suggest that people prefer or are comfortable doing shopping on their laptop or PC instead of their mobile. A good follow-up question to further understand the topic could be to ask the users which devices they use for online shopping.

Close to 80% of the people agreed they are unlikely to purchase if the website is not localized to their language. A fraction of the answers that disagreed with the statement, which is 19% might not have a problem purchasing on such a website, possibly if it is, i.e., available in the global language, which is English.

The survey results also suggest that users are paying attention to the loading time of the website and might not have patience if it has a slower performance.

When it comes to the contents of the website, half of the respondents admitted that they do not read large paragraphs of text. The statement was a reference to the study named "Text Scanning Patterns: Eye tracking Evidence" published by Nielsen Norman Group, which states that "People rarely read online — they are far more likely to scan than read word for word." (Pernice, 2019). Perhaps the weakness of this statement is the degree of subjectivity involved in what constitutes a 'large' paragraph. Some may consider a large paragraph as 300 words, others 100 words. Some may consider large paragraphs sources of information, others a cognitive burden. The question is based on research that tracks eye movements and

individual users' reactions to web content. The conclusion is that this kind of research should rely on live user data and not on user recall.

It was discovered that the negative emotions experienced during online shopping might stop the users from completing the purchase and lead to them not returning to this website anymore. More than 90% agreed and strongly agreed that the websites should be easy for them to use and provide the necessary information for them to consider it useful. This statement might also be considered subjective, but is still a good point for business owners and companies to consider.

The **Expert Survey**, which also included various UX principles in the statements, also suggests that it is essential for the user to feel safe and secure when browsing the website. Therefore, it is crucial to not only secure the website from the technical side but to provide the user with a sufficient amount of information to make sure he trusts the website to make a purchase.

Another useful finding that can be pointed out from the results is that it is essential to offer the information, features, and overall guidance in helping him to choose the products he needs. From the previous discussion about the e-commerce website elements, it is important to have a Search Bar present at all times.

When it comes to the Accessibility, the statement "All users of the website have the same experience, no matter what device they are using, the language they speak or any disability they may have" received both positive and negative responses from the e-commerce experts. One of the respondents made an effort to give feedback on the survey saying that "no matter how hard we try, a blind person will hardly have the same experience as a person without such a disability," which suggests that the website will be experienced differently by different users.

The statement that received the highest amount of positive feedback states that most of the experts, including e-commerce monetization experts, along with the consumer, strongly agree that it is essential for the website to be intuitive and easy to use. This idea of the website is continuously emphasized in the book by Steven Krug, named "Don't make me think!", which solely but its title suggests that there had to be as little friction when it comes to the online shopping as possible.

Since the primary goal of this thesis was to research the business value of UX and UI principles in E-commerce, the professionals mostly agreed to the statement that UX and UI design investment could promote increased sales and revenue, which are considered to be the leading indicators of success by many companies. This statement is supported by a strong expert agreement that investment in UX/UI design is quintessential and crucial for the business they are in.

The statement that received the lowest agreement rate among all was the claim that investment in UX and UI design can reduce the development cost and time in the long run, which received mixed answers. The answers suggest that it might not have an impact at all judging on the amount of "Neutral" responses.

Results strongly suggest that paying attention to UX principles is as important as monetization/marketing principles, such as pricing to the specific market and availability of multiple payment methods. Two areas of UX design were felt to be more important than pure monetization principles: 'website ease of use' and 'navigation to desired products'.

The **interview participants** agreed that UX and UI design help them in communicating better with the audience and users they are trying to sell their products to. Therefore, we can link that finding with the results of the statement from the survey, which suggests that investing in UX and UI design can lead to increased customer productivity, and received a close to 90% agreement rate.

One of the valuable findings from the research shows that investing and paying attention to the UX and UI design of the website can provide a competitive advantage and customer satisfaction, which was also confirmed by the interviewed experts with one of them stating that "Designers are really good in knowing what they can do to achieve the maximum customer satisfaction".

Interviewees strongly agreed that designers must rely on data to make sure that the best decisions are going to take place. The constant collaboration with all the parties involved is essential in order to be able to meet company goals and achieve positive results. Respondents agreed that no satisfactory results could be achieved if teams are not aligned with each other in terms of communication. Another thing that can be pointed out from the interviews is that all of the respondents are involved with e-commerce on different platforms and still share the same opinion on the topic.

6.2 Limitations of the thesis

It should be taken into account that the performed work on the thesis had certain limitations that could have affected the output presented in the previous chapters.

The first limit may be a lack of a sufficient amount of scientific and academic papers and previous research papers on the topic of User Experience and User Interface impact that is applied explicitly to E-commerce. This led to another limit, which was a high amount of grey literature used throughout the whole research. This might have affected the quality of the theoretical background given that when working with grey literature, it is difficult to determine its quality, find out whether the work was evaluated by someone or written by an expert in the field. Although there are many books published regarding User Experience and User Interface design, it was noticed that there is a low number of papers and theses on the popular scientific databases such as Web of Science and Scopus.

The next limitation can be considered to be the sample size of the quantitative research that was conducted, which was $n=189$ for the Consumer Survey and $n=26$ for Expert Survey. It is important to note that basing the study of a bigger sample size might have provided more accurate and diverse results. Although there were not found any significant differences in results when comparing different age groups and gender of respondents, it still can be

assumed that more of significant older population participation in the survey might provide more insightful results.

It should also be added that the experts that were willing to participate in the interview might have been, to some degree, biased towards the topic of discussion, which is usually the case with the type of research methodology chosen.

7 Conclusion

The bachelor thesis provided an advanced view on the topic of User Experience and User Interface from the E-commerce field perspective. The overall purpose of the thesis was to analyze the business value of UX/UI design and the importance of implementing individual principles in the digital commerce area.

The first goal was “To provide the theoretical background of E-commerce, User Experience, and User Interface areas” was realized in Chapter 1, which dealt with the definitions of each term and Chapter 2, where User Experience And User Interface were described in-depth using reliable sources on these specific topics.

The second goal “To provide an overview of the E-commerce website and its performance measurement” was realized in Chapter 3, which is focused on giving advanced information of what is an e-commerce website, the way it functions, a look at what a typical customer journey looks like and various methods on measuring the website performance from the UX point of view.

The third goal was focused on introducing various metrics for evaluating UX and UI performance, both of which can be used before and after any changes on the website.

The fourth goal was “To conduct and analyze surveys among the e-commerce consumers and experts, to interview the experts” involved gathering the data from typical online shopping consumers and e-commerce experts, which was successfully realized with the help of online surveys, the results of which were further visualized and analyzed. Additional data was also gathered during the interviews with e-commerce experts that accepted the request to provide their opinion based on years of experience. It can be confirmed that UX and UI principles can also be applied and can influence not only web-based but other e-commerce channels, such as mobile and desktop.

The overall results showcase the points that can be impacted by investing in the UX/UI design and the business value that it brings for the company, such as increased revenue and sales, increased customer productivity, satisfaction, and other cumulative effects in the long run. The results can also be used as a reference for companies that are considering implementing any of the UX strategies and would be interested in the opinions of other experts on the matter.

People use Internet services, applications, and e-shops, with the control and movement in them largely beginning to standardize. The last thing that makes a qualitative difference for the customer is the design.

In the future, it can be expected that the importance of customer experience, user experience, and visual design as a possible differentiator of business strategy will continue to grow. In a developed and highly competitive market such as E-commerce, continuous optimization across products, services, devices or communication channels with an

emphasis on experiences for organizations will increasingly no longer be a choice, but a strategic necessity.

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Annexes

Annex A: UX/UI influence points

The title of the source	Type of source	Author and/or publisher	Found number result
Clean, Cutting-Edge UI Design Cuts McAfee's Support Calls by 90%	Article	Bruce Hadley via pragmaticmarketing.com	3
Good UX Is Good Business: How To Reap Its Benefits	Article	Andrew Kucheriavy via Forbes.com	1, 3, 5, 6
The Trillion Dollar UX Problem: A Comprehensive Guide to the ROI of UX	Report	Multiple authors	1, 3, 4, 5, 6, 7
6 Significant Reasons to Invest in UX Design	Article	Maryna Z. via rubygarage.org	1, 2, 4, 6
User Experience (UX)	Article	Karyn Corrigan via oberlo.com	1
Top 8 Business Benefits Of User Research	Article	Paul Rouke via prwd.co.uk	1
eCommerce UX – measuring the business value of UX investments	Article	Marko Brisevac via inchoo.net	5
The business value of User Experience Design	Infographics	PointSource according to sources listed in the infographics	1, 2, 3, 5, 6, 7

Annex B: Interviews

Interview with Vojtech

Valeriya: Can you introduce yourself please and share what do you do?

Vojtech: I work as an Online Sales Manager for HMA VPN brand. So basically, I take care of the whole online monetization of HMA. And so that consists of acquisition. By that I mean first purchase monetization, retention, keeping users and customers with us. Checking the KPIs we have, forecasting as well and some project management of monetization projects. So generally, monetization and Checkout Cart projects. I have a team that this currently three people, including me.

So basically, our goal is, of course, to, you know, get sales and meet the plan. It's kind of tough for H.M.A. because the business is kind of mature and the competition is hard. I think we are doing great now. I'm part of broader team, which just about 10 people distributed in or located in the Czech Republic, London and Belgrade. I started this position almost a year ago and before I was doing utility business, first purchased and monetization. So now I'm having I make my own team with the acquisition and retention area. I work on our website and on the app that we're providing through the website.

My team and I are making sure that we have the right of us on the right channels, the right flows and everything is optimized. So we maximize the opportunity we have from the traffic that is being provided to us by channel teams. In the app we take care of messaging and expiration screens. But website is the main focus.

Valeriya: How closely does your team work with UX and UI designers? Describe the process?

Vojtech: Design is essential and we work very closely with all of the designers. Those are visual designers and UX designers. Generally, it is the face of the brand. So basically, we really need to attract the user. And we visualize what we offer in order to do so. And I think the visual design is essential and UX is also crucial for customer understanding.

So what we are trying to do is to make sure user understands what we are offering and that it may and it should actually meet the needs of the customer and of privacy. So generally speaking we have a very tight cooperation with the design team and we discuss every single button on a page in order to meet the sales expectations. And to make sure it resonates with customer.

Valeriya: Have you ever took part in the redesign or website optimisation project and what were the business objectives? How do you measure the success?

Vojtech: Sure. Redesign of the web usually relates to a bigger change either in a product or in a branding. So basically we did both. We did a redesign quite recently because the website was outdated. We had to make multiple changes on multiple places and it required a bigger change in how we communicate about the product. We also had a shift in branding and how

we present ourselves. We wanted to be a little more product oriented in terms of communication. But it was smooth. I think it took like surprisingly small time, around two months only. It was a good cooperation.

Regarding business goals, for monetization part it is always the Conversion Rate. So basically, we want to make sure that your conversion doesn't drop at least and we keep an eye on how users are converting and if there is any upside.

Valeriya: As an E-commerce expert, you can see the effect of any little changes that were made. Can you think of any example of the design change that has been made and what was the result of this change?

Vojtech: Definitely! Most of the time we use HotJar to see how users are interacting with our website and with that we have some knowledge about what we should do. I will show you the example.

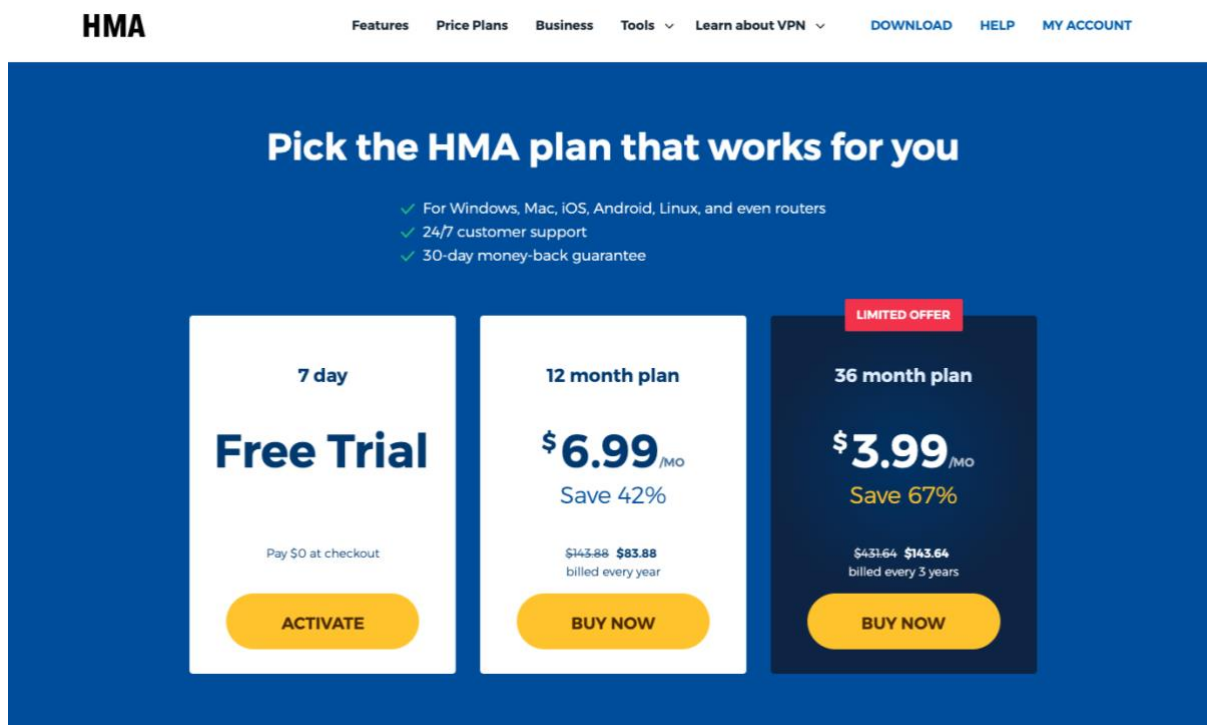


Figure 32: Interview attachment (Source: author's processing)

So recently, we completely changed the layout of our pricing page. It had a lot of unnecessary things in it that we removed and we also wanted to highlight the purchasing plan that is the best for us in terms of profitability, which is a 36 months. So yes, we did the UX and layout optimisation on this page that really improved the Conversion Rate.

We also have a couple of pop-up messages on the website to highlight the latest offers and deals. Our HMA logo is a donkey, so we use different variations of it to catch the eye.

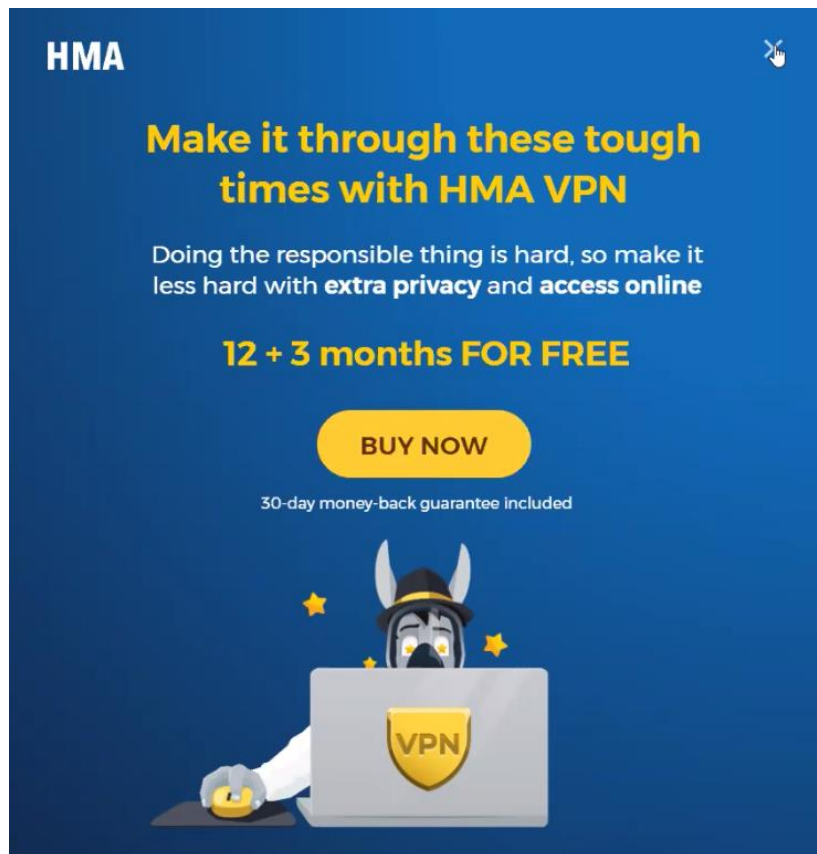


Figure 33: Interview attachment (Source: author's processing)

The Pajama Sale offer does not disappear when you close it but instead minimises, giving the user an option to come back to it if he decided that he is interested.

On the Home Page we have to make sure that we communicate the money-back guarantee so that we can be trusted and that alone has raised the conversion rate of this button up to another 5%. We can also create some urgency with the countdown of this limited offer. So basically all the small things were a part of redesign.

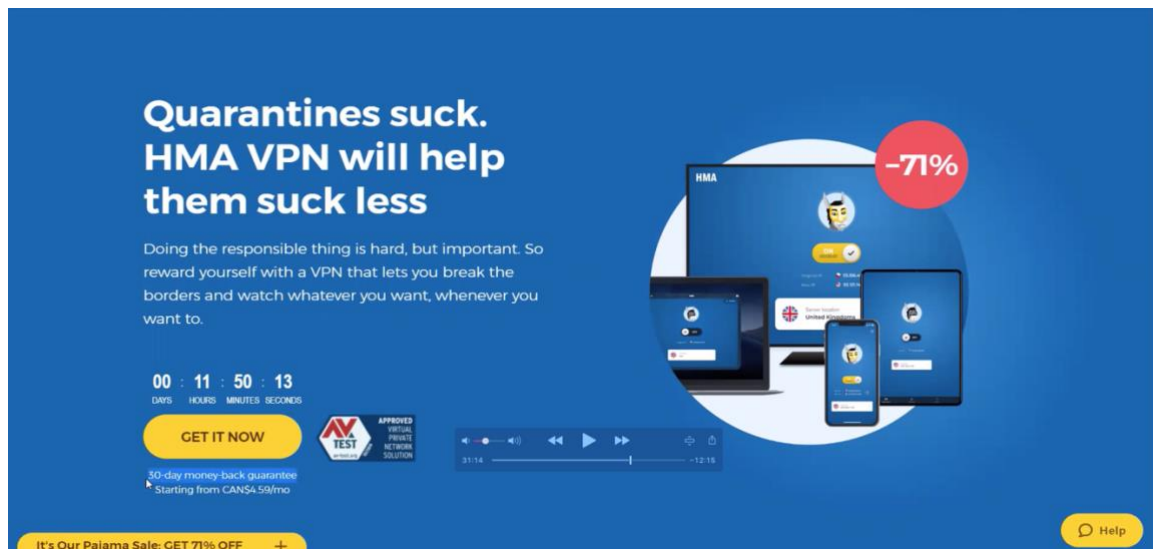


Figure 34: Interview attachment (Source: author's processing)

When measuring the success it usually really depends on what page we are talking about. But generally we typically measure RPM, which is Revenue per Mile. It means the revenue we are able to extract over thousand visitors. And if we do any change in the flows or the layouts or anything, you just observe how the RPM of the whole website, because it's ecosystem, how does it change. We are running A/B tests using Google Optimize to track everything. But RPM is the key indicator for us. We can also break down this RPM measure, it consists of Conversion Rate and AOV (Average Order Value). But the AOV, from my experience, doesn't change.

You can see the significant improvements from this report. We usually use Google's Data Studio to visualize the data. This report shows one of the bigger changes that we made for the website just recently, but you can see there is a huge growth in transaction count and page views (PV). The revenue per transaction is lower though, but it is nothing major and it happens sometimes. And the Conversion Rate is also changing during the tracked period, it's never still.

This is just one example but we definitely have thousands of similar ones from the past campaigns.

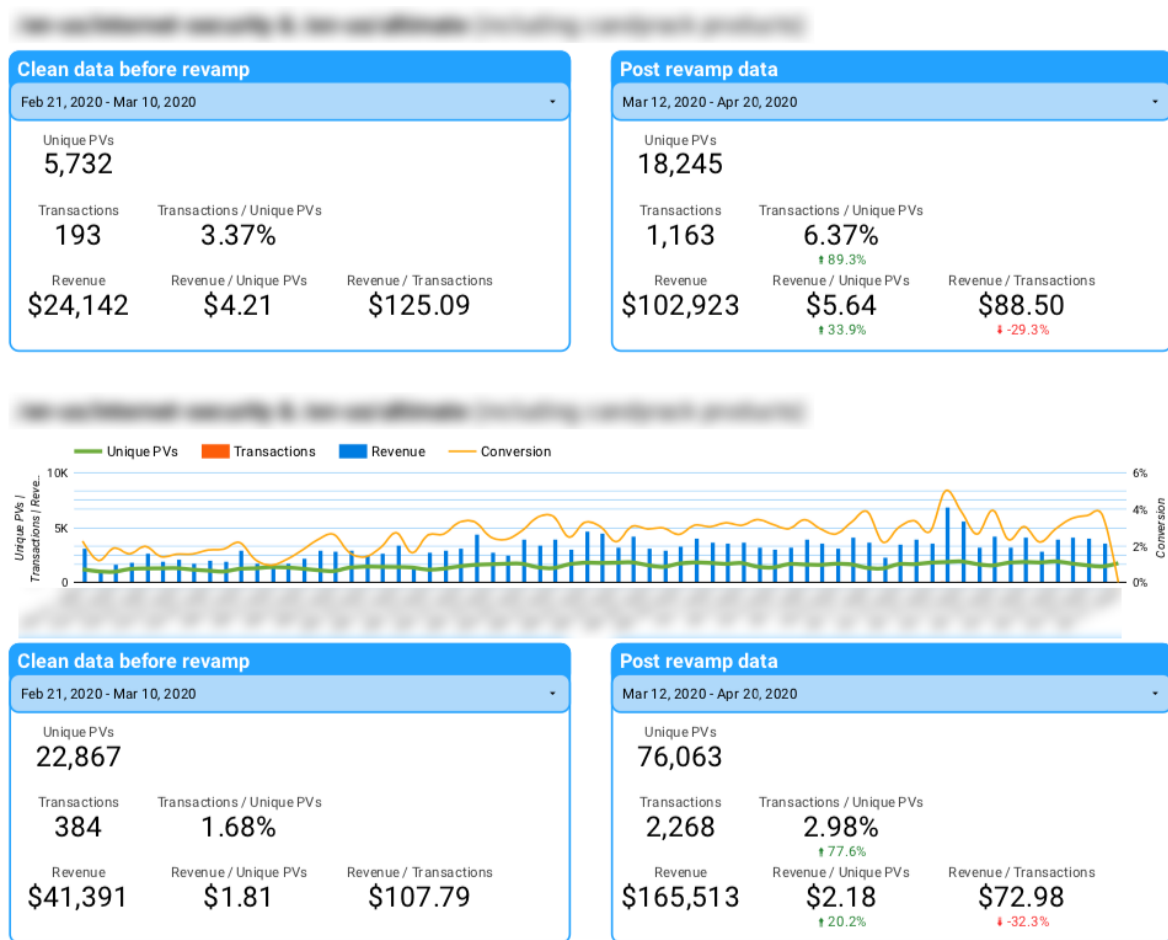


Figure 35: Interview attachment (Source: author's processing)

Note: the sensitive details were blurred out from this image upon the interviewees request in order to be able to include this information in the thesis.

Valeriya: Do you think that UX/UI design creates value for e-commerce business? Why?

Vojtech: Like I explained before, I think it is crucial. We improve the look and feel, we get increased sales and performance, we communicate and present out products in a better way. But it is very important that both design and Sales teams to understand each other. Sales and monetization people usually know that engagement of the user is the key for driving conversion and revenue. Designers are really good in knowing what they can do to achieve the maximum customer satisfaction, but they have to always be aligned with other teams to make sure we make profit. Overall, it is very important to keep an eye on both things and make sure we are going in the right direction.

1.1.3 Interview with Roman

Valeriya: Can you introduce yourself please and share what do you do?

Roman: I've been working in Avast for four years, before that I've been a consultant for a digital agency for two years comes them and and prior to that I've been a sales representative

for SMB in Hewlett Packard for four to five years. Overall I've been doing sales and sales related stuff. In Avast I am a Senior E-commerce Specialist and I'm working in the subscription team, which is a team and is dedicated to our paid subscriptions over desktop platform. I'm making sure that we retain our paid customers. Before this position I was a part of all consumer desktop acquisition team.

**Valeriya: How closely does your team work with UX and UI designers?
Describe the process?**

Roman: We work with them quite a lot. All the content we are displaying in this matter within the app, all of it need to go through UX designers, UI designers and copywriters before it reaches out developers. We are trying to optimize all of the sales flows constantly to provide better performance so I'm mostly involved with setting up the tests and trying to optimize whatever is possible to ensure that we reach our company goals. So whenever we have a new idea, something that can be a beneficial change to make to the flow, I contact designers and copywriters and they are bringing it to life.

Valeriya: Have you ever took part in the redesign or website/app optimisation project and what were the business objectives? How do you measure the success?

Roman: Of course. We do them quite often. I will make a very simple example. So imagine that your app subscription is expiring. And we need to let you know and we can send you just a simple email to notify you. But we also know that email opening rate is about 20%. We also know that you are for sure using our app on the daily basis, let's say. So we send you the message within the app itself.

But let's say we have an idea. For example we want to change the CTA (call-to-action) button on the screen that says "Your subscription expires in five days". The button might now say 'Renew' and we want to change it to saying "Resubscribe". Or we can have something else in mind. But before we make any actual changes we need to do some A/B testing comparing these two options. If the changes were already made we might look at the number on clicks change and the RPM measure to tell whether the change was successful or not.

And I'm basically then responsible for the evaluation of the whole task, analyzing the results and if the results were positive I'm responsible for the delivery. We usually look at the data and our main focus is of course to boost sales.

Valeriya: From the survey answers that you submitted I saw that you highly agreed to the statement which said that investing into UX and UI website design can reduce development costs in the long run? Can you explain why you chose to agree on this?

Roman: I highly agreed because if you already have a good UX and you can provide your customers with a good experience then you're saving money in the long run. In my opinion, if you have a start-up or a small business and you have money I would consider it to be number one thing to consider.

Frankly speaking, my opinions on UX differ quite a lot when talking about different platforms. Let's say from my experience the impact of UX on sales profitability is completely different on Web, on applications and other sales channels.

In Avast most of our revenue is coming from our app and from my experience, one of the most important things when it comes to conversion of our users is whether these users engage with the app, so in this sense paying attention to UX is completely quintessential for us to be profitable, which is similar to the web but from my experience it much more important on web. So generally, if people do not use our product or are not satisfied with it they are unlikely to convert from the sales flows that are coming from there.

Valeriya: Can you think of any example when you made any changes to the app design and what was the effect of that change? Does it have any impact on conversion?

Roman: This is a tricky question. From my professional experience, I've seen the changes in Conversion rate uplift ranging from 5%, which to me is a flat result, up to 20%.

We've seen a lot of effects after changing UX and UI of the app, positive and negative. The only that I constantly talk about is that designers are not very sales oriented people. It is very important to analyse the data to make sure that you make correct decisions. And we typically receive positive results only if the Sales department and Design department are collaborating properly. Because when they are not the results are not at all satisfactory. Everything in the process should be intertwined and one can't be done without the other.

Valeriya: Do you think that UX/UI design creates value for e-commerce business? Why?

Roman: It for sure does, but as I said, it is good to rely on data. You have to be able to read numbers in order to do correct decisions. I also want to note that it doesn't bring the never-ending value, if that makes sense. Even the team with the best designers in the world can't produce constant growth as the exact pace. If we were to wake up everyday with some groundbreaking ideas we would probably be multi-billionaires by now. There is an edge and peak of everything and it's impossible to always have this continuous growth. And the progress is usually steady and low, you can only see it over time.

Valeriya: Something else you want to add?

Roman: Yes! I think that there is a constant pressure to deliver value. What I mean is that the Sales people create a lot of pressure on to other teams, including design team, to deliver the value for them. And that's not always the best way how to treat the creative people. I saw that in a lot of other companies and projects and so on. And I'm not sure whether that's something that can be easily addressed or resolved.

1.1.4 Interview with Petr

Valeriya: Can you introduce yourself please and share what do you do?

Petr: My name is Petr and basically I am taking care of the privacy mobile segment in Avast. My role is a Senior E-commerce Specialist. The department is basically consumer oriented. It basically means that we take care of the online monetization of our apps. One part is the desktop. The second part is the mobile. I work in the mobile department. And how I already mentioned like my role is like we care about the privacy segment, those are VPN and other privacy products.

**Valeriya: How closely does your team work with UX and UI designers?
Describe the process?**

Petr: Very closely. Basically, for us, it's essential to create any sales activity. So basically we go through the circle that includes copywriters, the designers both UX and UI. It's always depends on what kind of activity it is. Then it goes to localization team. And then it goes to the technical side where everything is getting set up based on our guidance. But overall, every single activity that is being done, we make sure to always include the design team.

Valeriya: Have you ever took part in the redesign or website/app optimisation project and what were the business objectives? How do you measure the success?

Yes. We definitely are forced to constantly innovate and try to bring new things to the table. So a part of our job is constantly testing. It can be the Home screen, an overlay, it can be notifications. It is like a lot of stuff that we can test. So basically we are revamping or trying new stuff every single month. So it's not like we get to, let's say, a certain point and we I say "OK, that's working. Let's wait another year!". We are constantly innovating and and trying new stuff and new design approaches. It is important for us to never stay still and find different ways to improve what we currently have and make it even better. We are preparing the new release of the app that we put together along with designers and we definitely take the monetisation part into account.

We come up with some side suggestions and then we are brainstorming on how we can handle this and that, how we can incorporate this kind of offer and iterate till we get the final status where we are very comfortable with how we are able to walk the customer and hit him several sales touchpoints to be sure that user and ensure that he will eventually make a purchase. Or at least try our product. So basically we are designing a whole funnel to make sure that user understands how the product works. We make sure that we hit him several times with, sales activity because at the end it's all about the revenue and we have to meet company goals.

Valeriya: What tests do you usually do before release?

Petr: In terms of new product, basically there's something called soft launch. So basically decide to choose one country, which for us is the control group, for example, for the entree check. It will be the Australia, the market is quite huge. And also, oh, the customers are behaving very similar to US and Canada, that are also big markets. We if we know that it's working well in Australia then it will most probably work in other countries.

Then we are observing how our users are behaving. How many users install the app, how much time do they spend on it, what offer did they purchase first. And based on that data

we can do optimisation. If we see that 80% of users leave without going any further into the app then something is probably wrong with the screen they see for the first time. Or maybe there are some technical issues.

We look at RPM measure, which is Revenue per thousand people. If we do any changes to CTA button and the RMP metrics value increases, then we know that we're on the right track. We also have RPI metrics, which is revenue per install, but this one is very app specific.

Generally, there are thousands of different metrics that we look on including Conversion Rate, the Average Order Value and so on.

Valeriya: As an E-commerce expert you can see the effect of any little changes that were made. Can you think of any example of the design change that has been made and what was the result of this change?

Petr: From my experience, I've seen very little and subtle changes that had a huge impact. We are always trying to emphasise what the deal for the user that will bring us the biggest profit. Even if it's a small change in the way we communicate our prices make a huge difference. I can't share the exact numbers and data but I can confirm that it makes quite an impact on multiple metrics including Conversion Rate of course.

Valeriya: Do you think that UX/UI design creates value for e-commerce business? Why?

It definitely does. For example here in Avast, the design is something that creates a brand. A user has to know that this page or this screen are from Avast. And based on the research we know that branding is an extremely powerful thing and you have to emphasise it with the interface. Like I mentioned many times, User Experience is crucial and from my experience in the previous companies, it is something that needs to be taken care of.

E-commerce is a very dynamic environment, you can leave the app or the website in a matter of half a second and you're gone. And you never even remember where you were a second ago. The duo of UX and UI is what ultimately creates a good customer experience, which supports sales. I don't think that we would be able to reach our company goals and objectives without investing into these two areas. And even if we received some results, they would only be short term, because what creates long-term value is being customer oriented.

Design and sales department have a very special relationship. Designer might provide something that is very good in terms of customer satisfaction, but might not bring any monetary value. There is a lot of back and forth discussions between the data team, sales team and design. But in the end they come up with the solution that satisfies all the parties involved.